

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7280

Nombre de lignes df clean	7280
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5096
Taille dataset test	2184
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

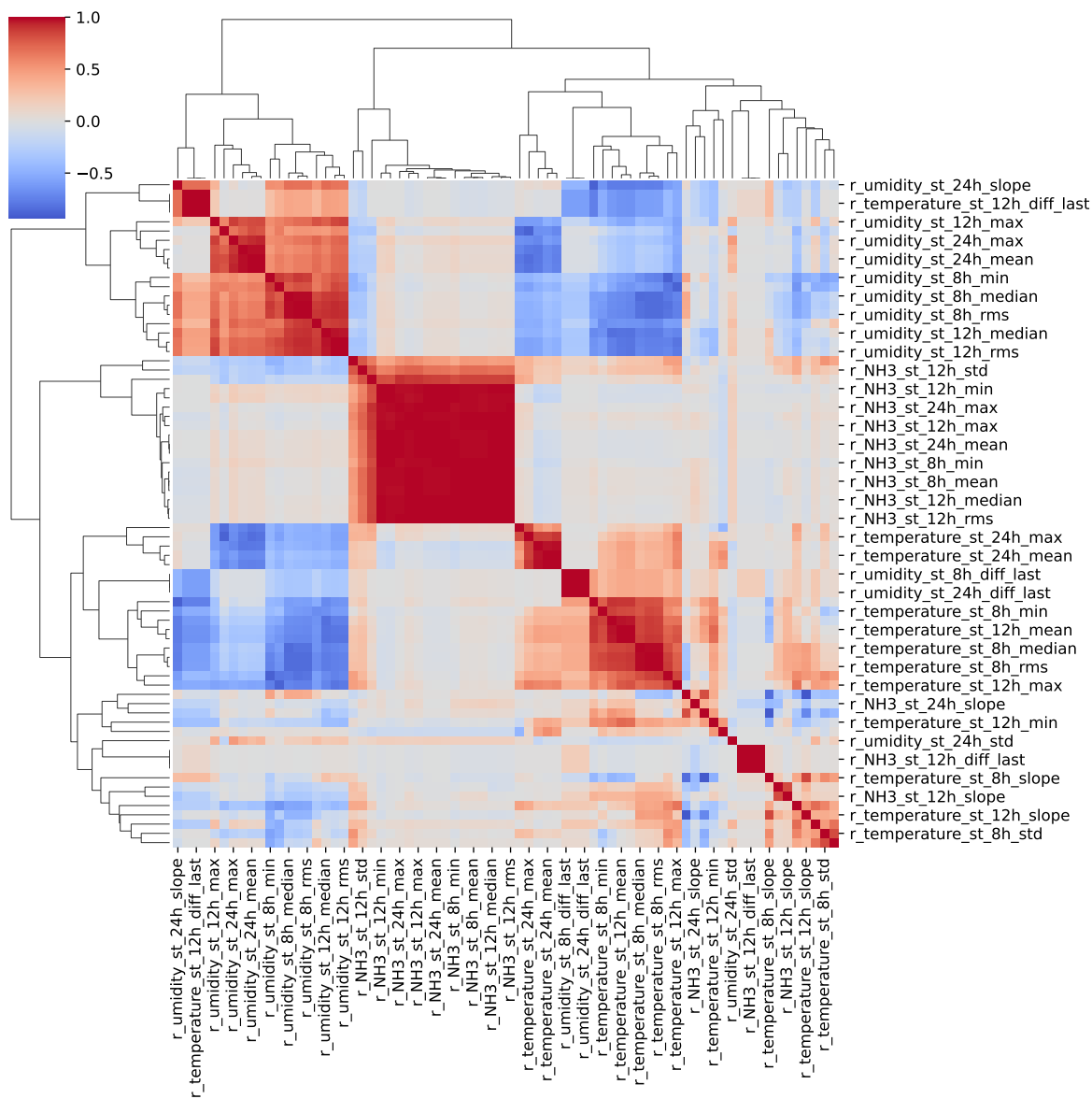
cluster 1: r , N , H , β ; **cluster 2:** r , t , e , m , p , e , r , a , t , u , r , e ; **cluster 3:** r , u , m , i , d , i , t , y ; **cluster 4:** r , C , O , 2 ; **cluster 5:** r , T , H , I ; **cluster 6:** $r_NH3_st_12h_min$, $r_NH3_st_8h_rms$, $r_NH3_st_8h_min$, $r_NH3_st_12h_mean$, $r_NH3_st_8h_median$, $r_NH3_st_12h_rms$, $r_NH3_st_24h_min$, $r_NH3_st_24h_max$, $r_NH3_st_12h_median$, $r_NH3_st_8h_mean$, $r_NH3_st_24h_rms$, $r_NH3_st_24h_median$, $r_NH3_st_8h_max$, $r_NH3_st_12h_max$, $r_NH3_st_12h_std$, $r_NH3_st_24h_std$, $r_NH3_st_24h_mean$; **cluster 7:** $r_NH3_st_8h_std$; **cluster 8:** $r_NH3_st_8h_diff_last$, $r_NH3_st_12h_diff_last$, $r_NH3_st_24h_diff_last$; **cluster 9:** $r_NH3_st_12h_slope$, $r_NH3_st_8h_slope$; **cluster 10:** $r_NH3_st_24h_slope$; **cluster 11:** $r_umidity_st_12h_rms$, $r_temperature_st_12h_rms$, $r_temperature_st_8h_rms$, $r_umidity_st_12h_mean$, $r_umidity_st_8h_max$, $r_temperature_st_24h_rms$, $r_temperature_st_24h_mean$, $r_temperature_st_12h_mean$, $r_umidity_st_8h_median$, $r_umidity_st_24h_median$, $r_temperature_st_8h_mean$, $r_temperature_st_24h_median$, $r_temperature_st_8h_min$, $r_temperature_st_8h_max$, $r_umidity_st_8h_mean$, $r_umidity_st_12h_max$, $r_umidity_st_24h_rms$, $r_umidity_st_12h_median$, $r_umidity_st_24h_slope$, $r_temperature_st_24h_std$, $r_umidity_st_8h_min$, $r_umidity_st_24h_mean$, $r_temperature_st_12h_max$, $r_umidity_st_8h_rms$, $r_umidity_st_12h_min$, $r_temperature_st_24h_slope$, $r_umidity_st_24h_max$, $r_temperature_st_12h_median$, $r_temperature_st_8h_median$, $r_umidity_st_24h_min$, $r_temperature_st_24h_max$; **cluster 12:** $r_temperature_st_8h_std$, $r_umidity_st_8h_std$; **cluster 13:** $r_temperature_st_12h_diff_last$, $r_temperature_st_8h_diff_last$, $r_temperature_st_24h_diff_last$; **cluster 14:** $r_temperature_st_8h_slope$, $r_umidity_st_12h_slope$, $r_umidity_st_8h_slope$, $r_temperature_st_12h_slope$; **cluster 15:** $r_temperature_st_12h_std$; **cluster 16:** $r_temperature_st_12h_min$; **cluster 17:** $r_temperature_st_24h_min$; **cluster 18:** $r_umidity_st_8h_diff_last$, $r_umidity_st_12h_diff_last$, $r_umidity_st_24h_diff_last$; **cluster 19:** $r_umidity_st_12h_std$; **cluster 20:** $r_umidity_st_24h_std$;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3 , $r_temperature$, $r_umidity$, r_CO2 , r_THI , $r_NH3_st_12h_min$, $r_NH3_st_8h_std$, $r_NH3_st_8h_diff_last$, $r_NH3_st_12h_slope$, $r_NH3_st_24h_slope$, $r_umidity_st_12h_rms$, $r_temperature_st_8h_std$, $r_temperature_st_12h_diff_last$, $r_temperature_st_8h_slope$, $r_temperature_st_12h_std$, $r_temperature_st_12h_min$,

r_temperature_st_24h_min, r_umidity_st_8h_diff_last, r_umidity_st_12h_std,
r_umidity_st_24h_std



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_12h_min, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_slope, r_umidity_st_12h_rms, r_temperature_st_8h_std, r_temperature_st_12h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_24h_min, r_umidity_st_8h_diff_last, r_umidity_st_12h_std, r_umidity_st_24h_std

Len(Features_X): 20

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

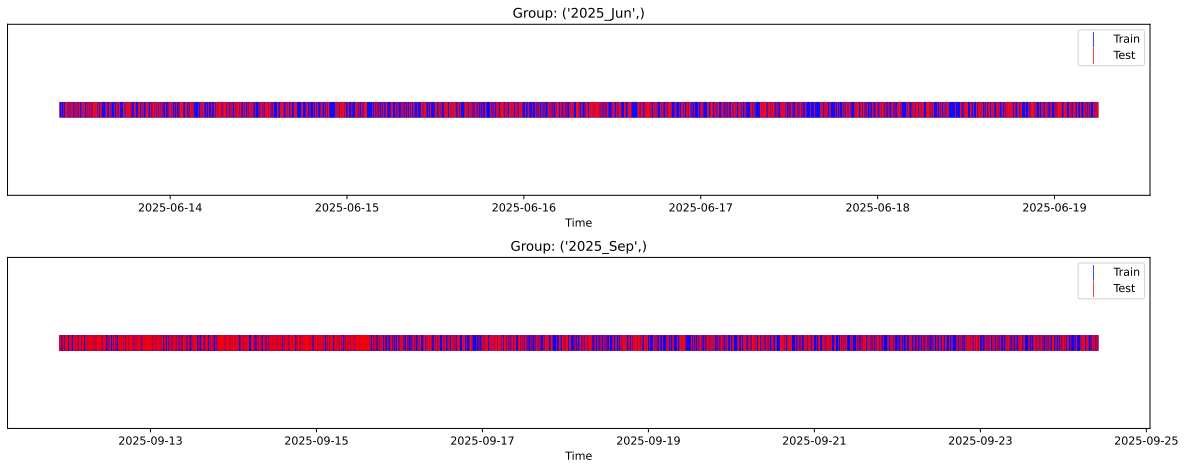
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

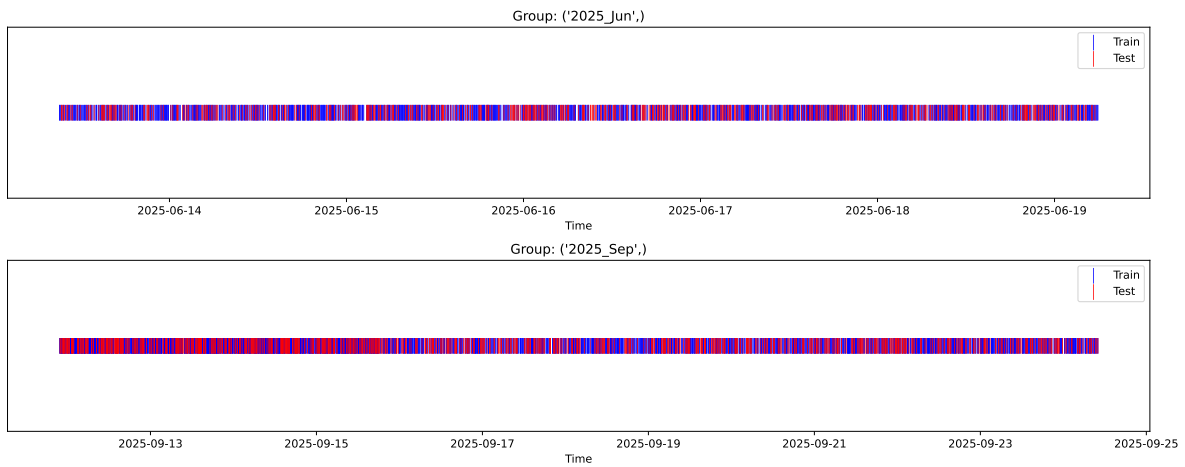
External Split: Train: 5096 samples Test: 2184 samples



3.3.2 Internal CV Folds

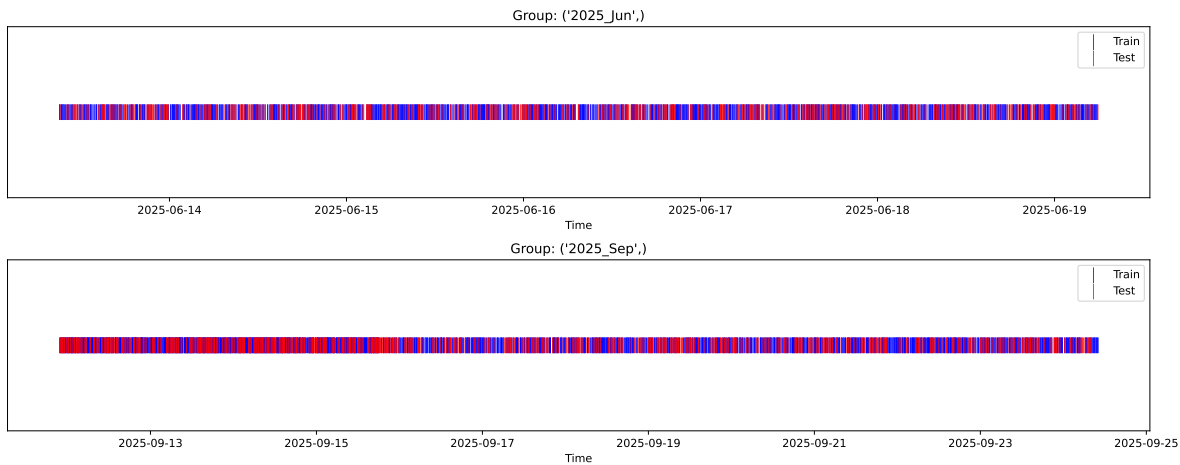
Fold 1/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



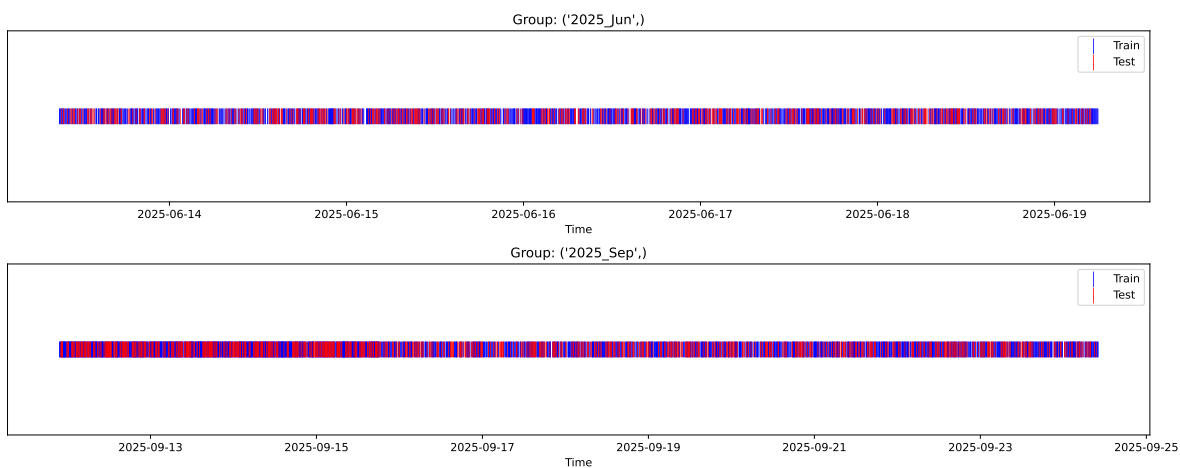
Fold 2/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



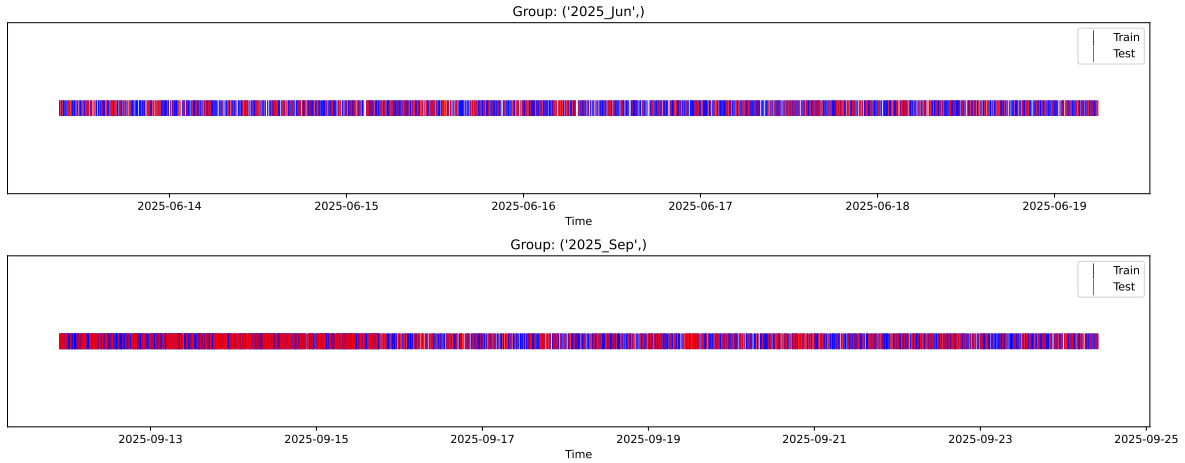
Fold 3/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



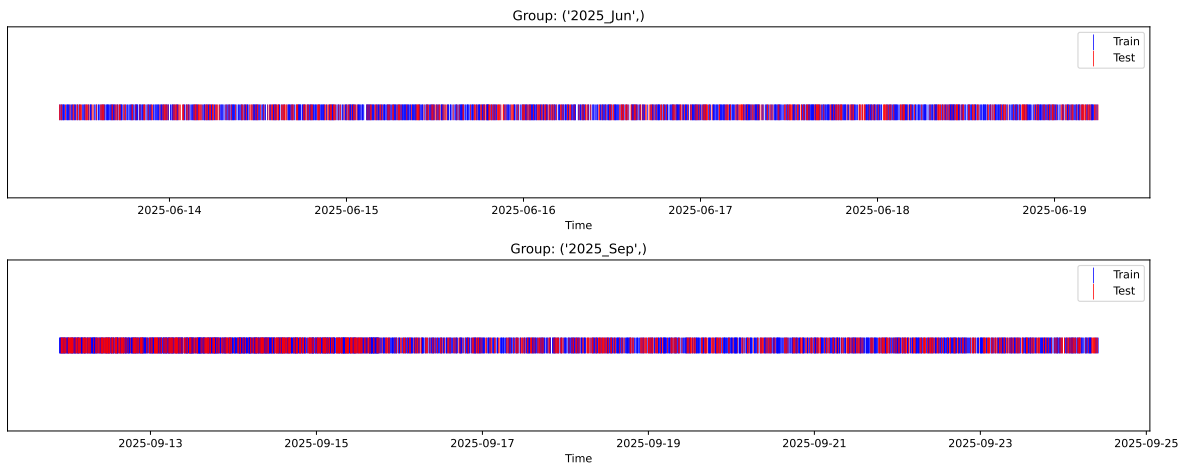
Fold 4/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.362, std=0.816, min=0.959, max=13.845
 y_{test} : mean=2.359, std=0.781, min=1.062, max=7.822

Fold 0 — y_{test} : mean=2.36, std=0.87, y_{train} : mean=2.36, std=0.79

Fold 1 — y_{test} : mean=2.38, std=0.84, y_{train} : mean=2.36, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.83, y_{train} : mean=2.36, std=0.81

Fold 3 — y_test: mean=2.36, std=0.81, y_train: mean=2.36, std=0.82

Fold 4 — y_test: mean=2.39, std=0.85, y_train: mean=2.35, std=0.80

train scores CV: [0.57501258 0.56762627 0.56033955 0.53530684 0.56441611]

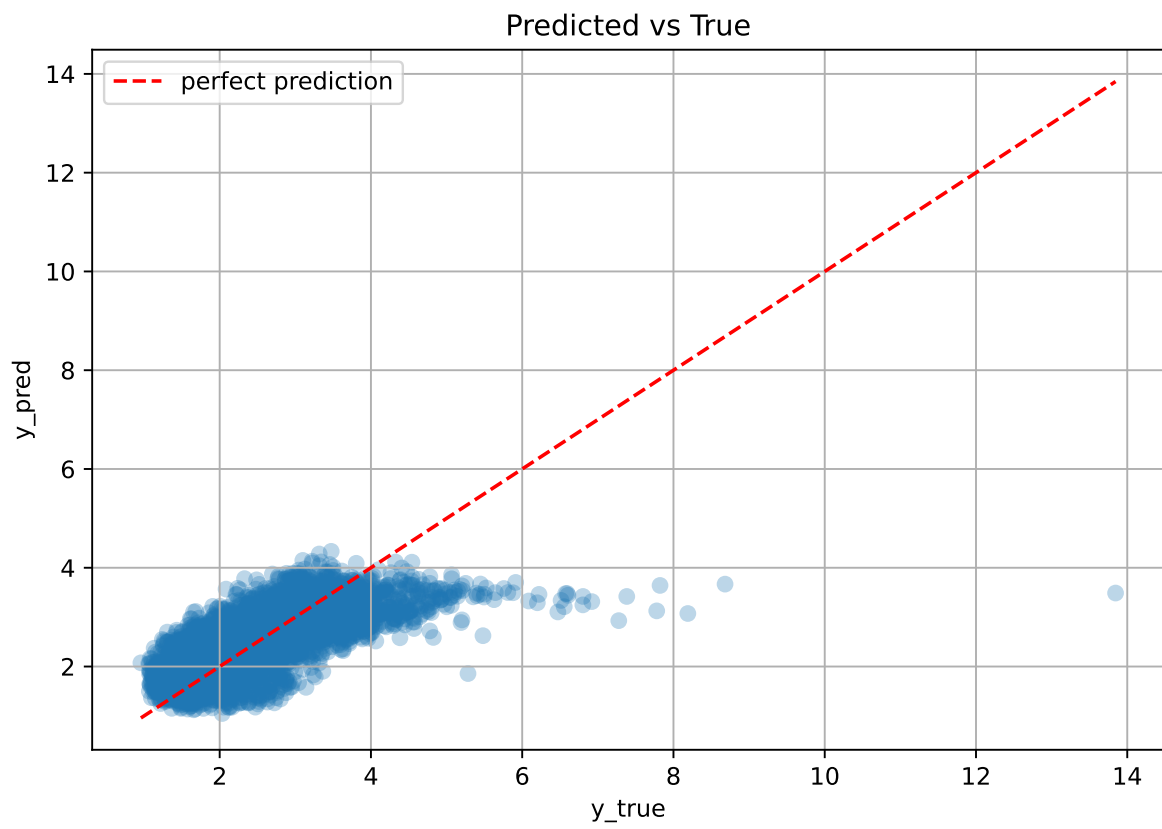
test scores CV: [0.47855796 0.48951682 0.50289486 0.56012104 0.4963227]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5605	0.5055	0.0285	0.5649	0.566	0.0605	1.1198	1.1089

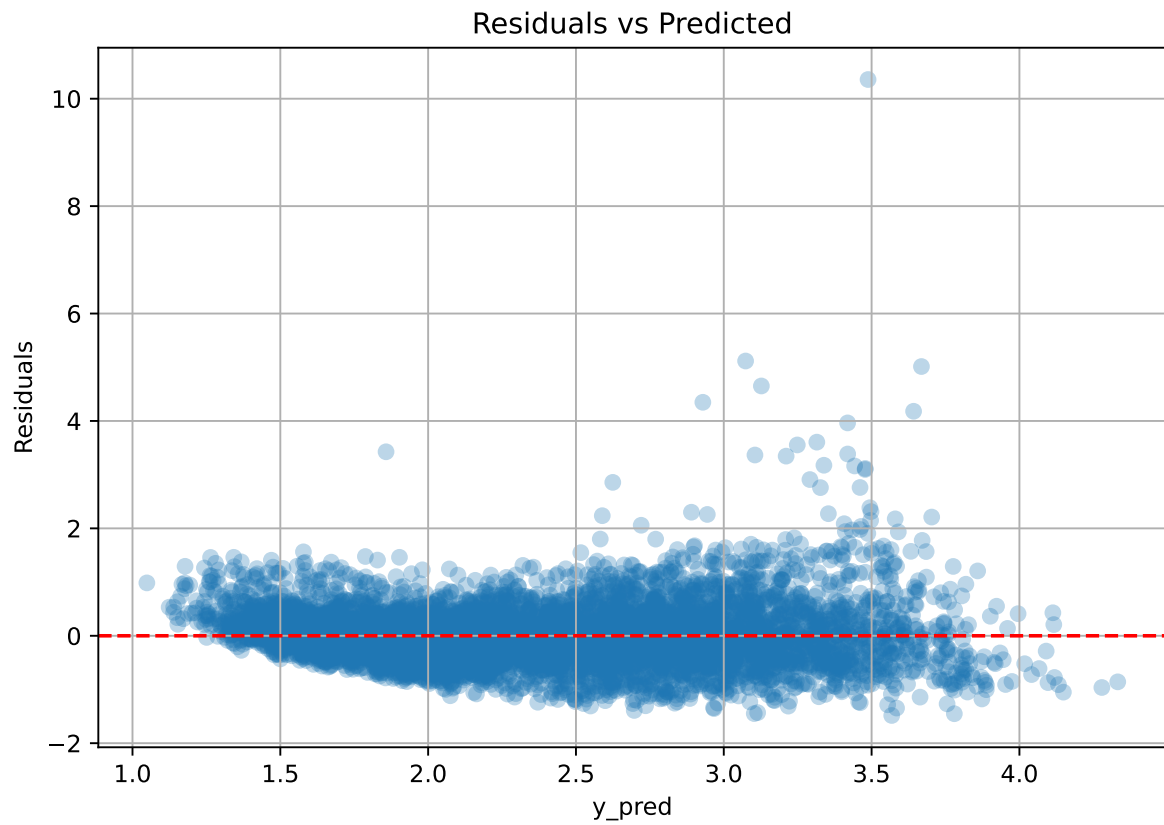
Coefficients:

coef_r_CO2 = -0.0576, coef_r_NH3 = 0.4771, coef_r_NH3_st_12h_min = -0.5459, coef_r_NH3_st_12h_slope = -0.0853, coef_r_NH3_st_24h_slope = 0.0386, coef_r_NH3_st_8h_diff_last = 0.0052, coef_r_NH3_st_8h_std = 0.0279, coef_r_THI = 0.5156, coef_r_temperature = -0.3787, coef_r_temperature_st_12h_diff_last = 0.1398, coef_r_temperature_st_12h_min = -0.0153, coef_r_temperature_st_12h_std = -0.0839, coef_r_temperature_st_24h_min = -0.0557, coef_r_temperature_st_8h_slope = -0.2001, coef_r_temperature_st_8h_std = 0.0752, coef_r_umidity = -0.5923, coef_r_umidity_st_12h_rms = 0.1562, coef_r_umidity_st_12h_std = -0.1849, coef_r_umidity_st_24h_std = 0.0285, coef_r_umidity_st_8h_diff_last = -0.0721, intercept = 2.3622,

3.4.2 Scatter



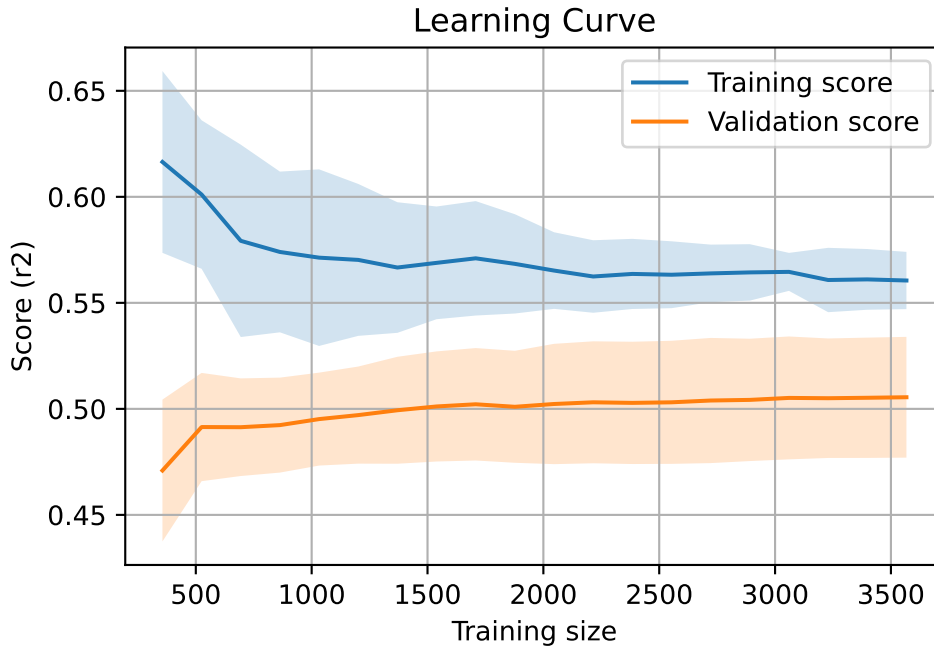
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7280, **Len(training)** : 5096,

Len(training_real) : 3568, **Len(test_of_training)** : 1528, **Len(test)** : 2184



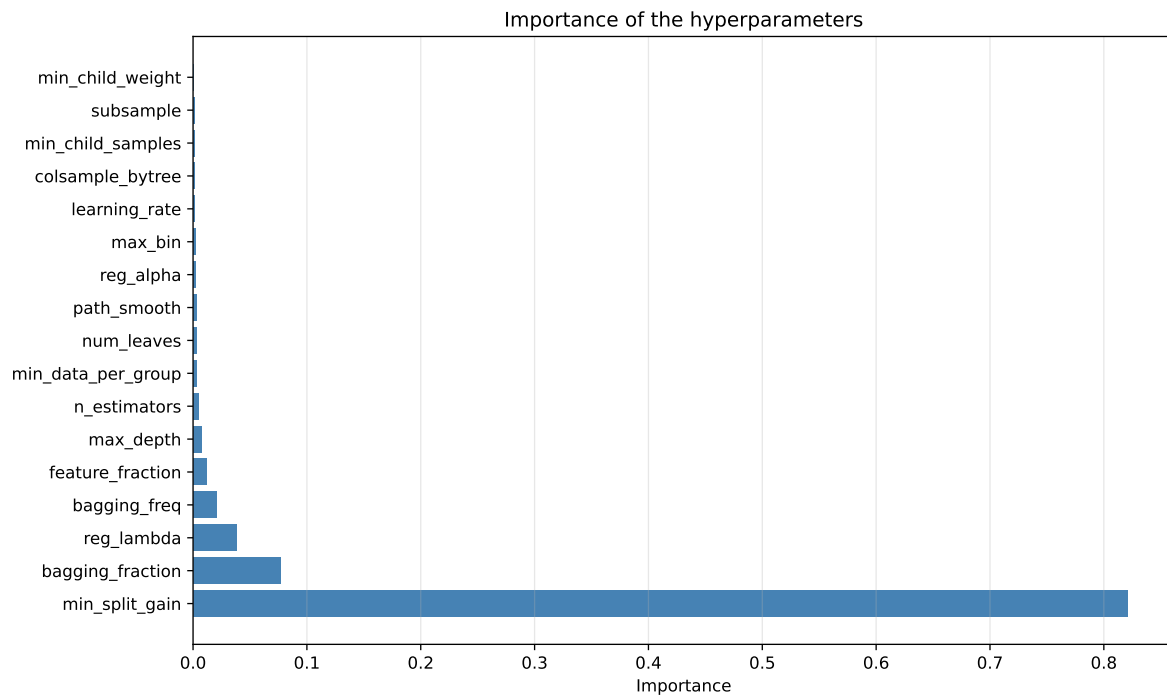
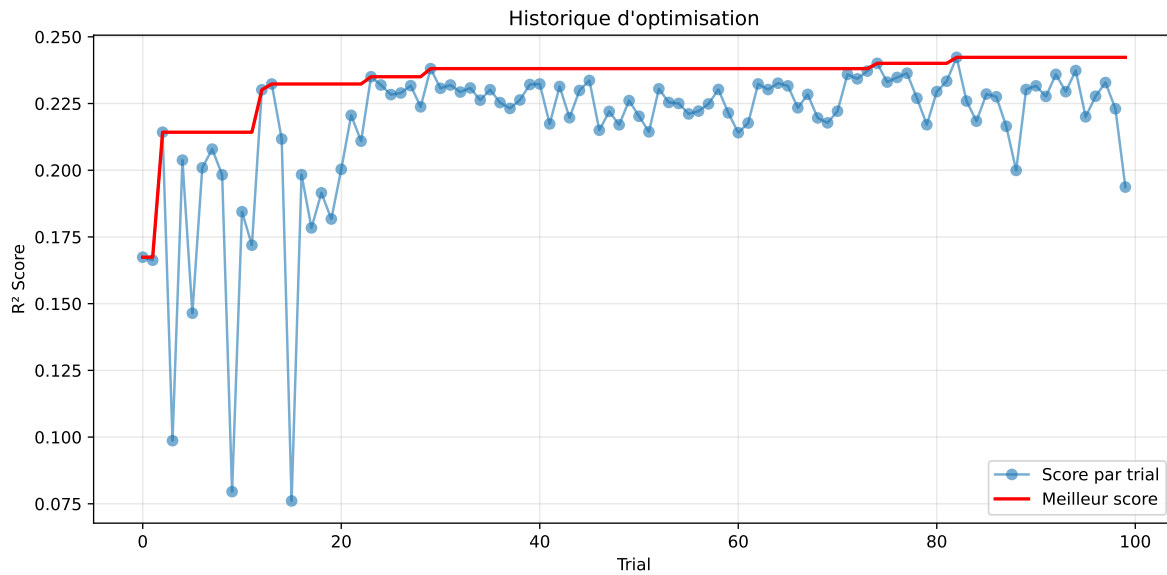
3.5 Model : LGBM

3.5.1 Gridsearch

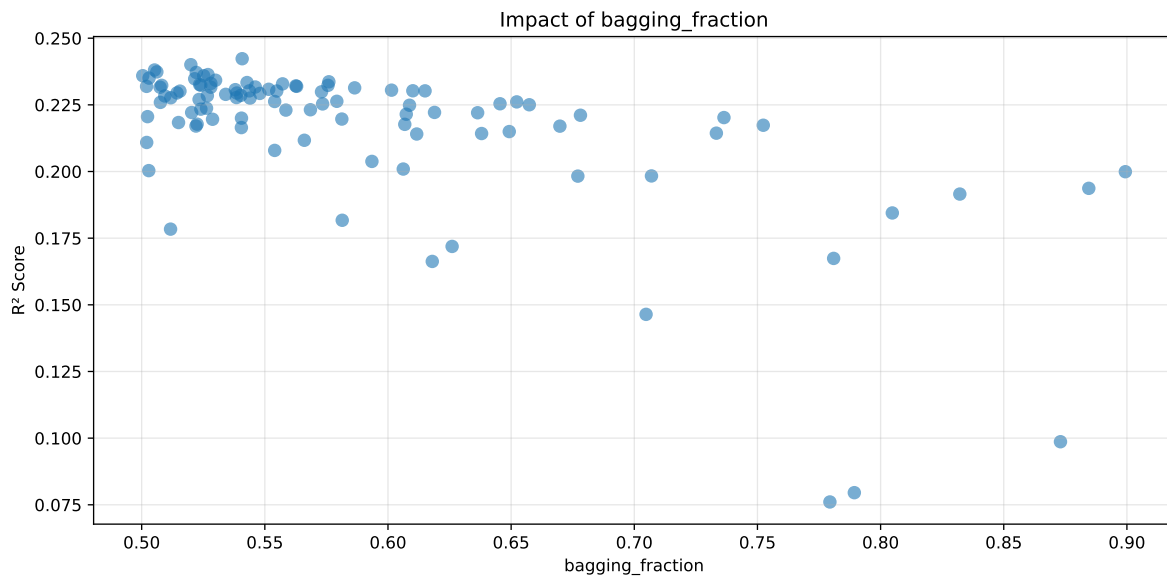
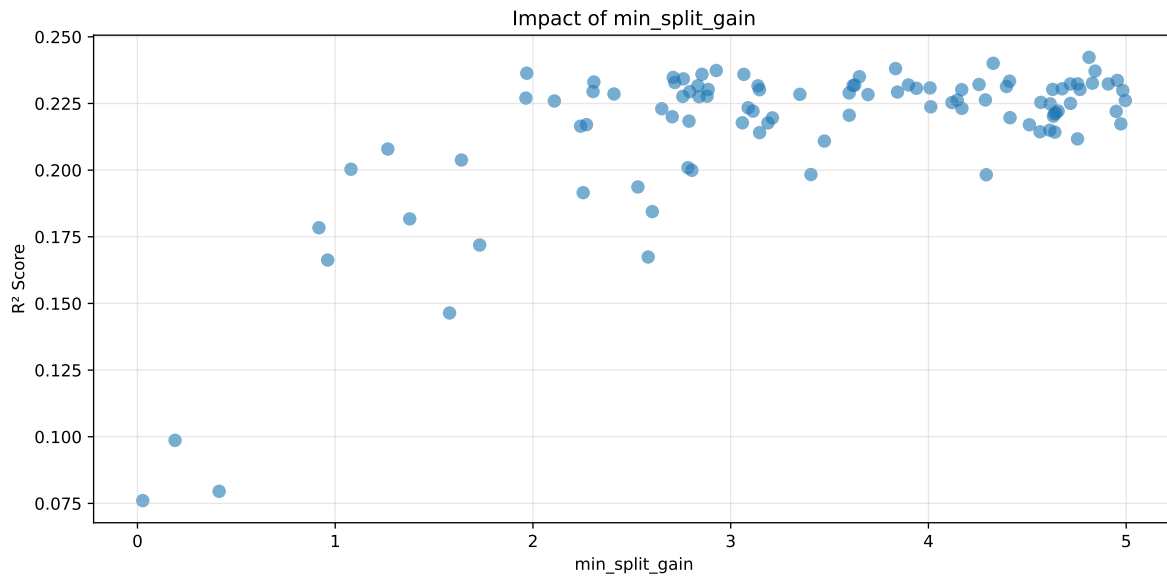
Distribution y_{train} vs y_{test} : y_{train} : mean=2.362, std=0.816, min=0.959, max=13.845
 y_{test} : mean=2.359, std=0.781, min=1.062, max=7.822

===== TOP Importance FEATURES =====

feature	importance
r_umidity	68
r_temperature_st_12h_diff_last	56
r_temperature_st_24h_min	45
r_umidity_st_12h_std	35
r_temperature	33
r_temperature_st_8h_slope	30
r_CO2	27
r_NH3_st_12h_min	26
r_umidity_st_12h_rms	20
r_umidity_st_8h_diff_last	20
r_NH3_st_8h_diff_last	11
r_temperature_st_12h_std	10
r_NH3_st_8h_std	9
r_THI	7
r_umidity_st_24h_std	5
r_NH3_st_12h_slope	5
r_temperature_st_8h_std	1
r_NH3_st_24h_slope	1
r_temperature_st_12h_min	0
r_NH3	0

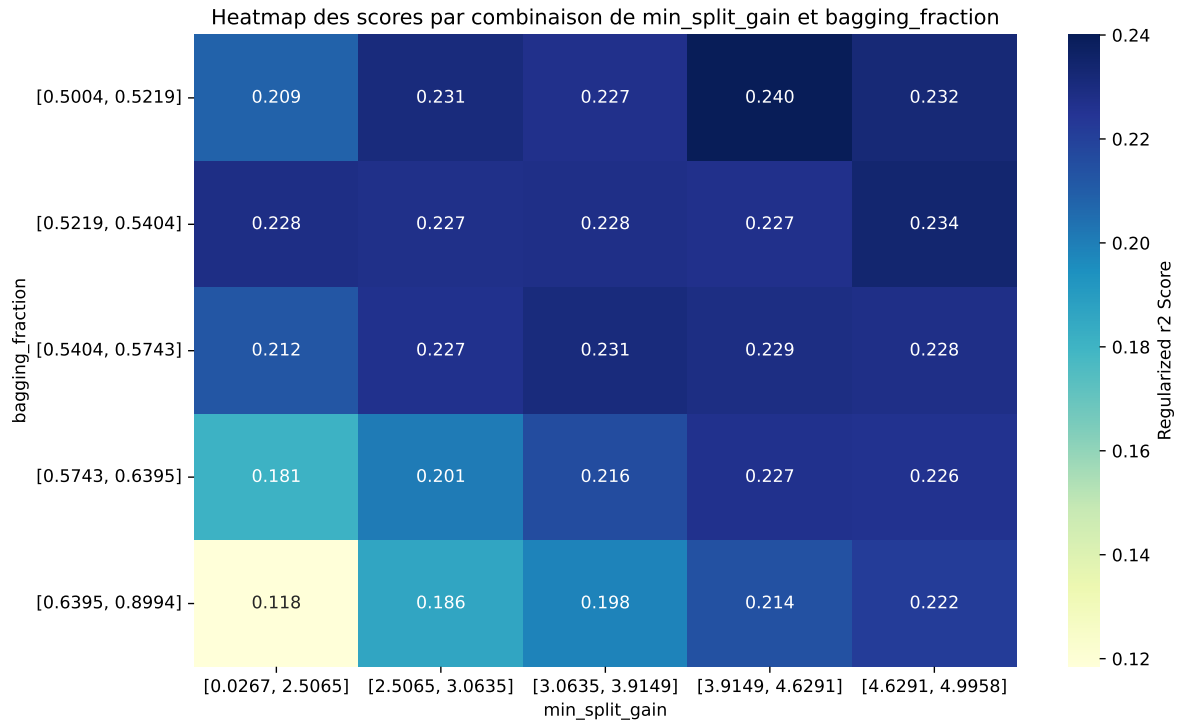


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6144	0.5524	0.6238	0.0328	0.2423	0.6483	0.0959	1.1736	1.1123
2	0.609	0.5475	0.6194	0.032	0.2401	0.6408	0.0933	1.1705	1.1123
3	0.6252	0.5606	0.6334	0.0301	0.2381	0.6574	0.0967	1.1725	1.1151
4	0.6341	0.5679	0.6393	0.0269	0.2374	0.6697	0.1017	1.1791	1.1164
5	0.6066	0.5451	0.6175	0.0321	0.2372	0.6302	0.0851	1.1561	1.113
6	0.6471	0.5786	0.6504	0.0301	0.2364	0.68	0.1014	1.1752	1.1183
7	0.6555	0.5856	0.6585	0.0297	0.2359	0.6735	0.0879	1.1501	1.1194
8	0.6338	0.5675	0.6387	0.0325	0.2359	0.6692	0.1018	1.1793	1.1169
9	0.6159	0.5524	0.6249	0.0299	0.235	0.6436	0.0912	1.1652	1.1149
10	0.633	0.5666	0.6388	0.0254	0.2348	0.6629	0.0962	1.1698	1.1171
---	---	---	---	---	---	---	---	---	---
100	0.8468	0.7184	0.7929	0.0304	0.0761	0.8158	0.0974	1.1356	1.1788

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.043183066338015906, 'reg__num_leaves': 53, 'reg__subsample': 0.5976974104348702, 'reg__colsample_bytree': 0.5355616888940967, 'reg__min_child_samples': 25, 'reg__reg_alpha': 7.630229837793921, 'reg__reg_lambda': 47.746068282842536, 'reg__min_split_gain': 4.812150399714436, 'reg__path_smooth': 0.11127543948940732, 'reg__min_child_weight':

0.32943581697610047, 'reg__feature_fraction': 0.7373612157764402, 'reg__bagging_fraction': 0.5407996566055822, 'reg__bagging_freq': 4, 'reg__max_bin': 73, 'reg__min_data_per_group': 155}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.04183763993976362, 'reg__num_leaves': 16, 'reg__subsample': 0.6377455220920428, 'reg__colsample_bytree': 0.529068589444384, 'reg__min_child_samples': 33, 'reg__reg_alpha': 9.436016478385564, 'reg__reg_lambda': 48.21966788653859, 'reg__min_split_gain': 4.327463879860011, 'reg__path_smooth': 2.4580849375427754, 'reg__min_child_weight': 0.49533828045344114, 'reg__feature_fraction': 0.7381247485322562, 'reg__bagging_fraction': 0.5199660586846991, 'reg__bagging_freq': 4, 'reg__max_bin': 76, 'reg__min_data_per_group': 157}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.02585546274873268, 'reg__num_leaves': 52, 'reg__subsample': 0.6573587632356437, 'reg__colsample_bytree': 0.5256508523848651, 'reg__min_child_samples': 10, 'reg__reg_alpha': 5.657247612111149, 'reg__reg_lambda': 48.02422671401094, 'reg__min_split_gain': 3.83361773911994, 'reg__path_smooth': 1.5194869920160452, 'reg__min_child_weight': 0.022334303205117267, 'reg__feature_fraction': 0.720069000428208, 'reg__bagging_fraction': 0.5052347735169966, 'reg__bagging_freq': 6, 'reg__max_bin': 64, 'reg__min_data_per_group': 147}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.049377521260757874, 'reg__num_leaves': 18, 'reg__subsample': 0.8425627470302024, 'reg__colsample_bytree': 0.6313151046005068, 'reg__min_child_samples': 38, 'reg__reg_alpha': 7.476534602685021, 'reg__reg_lambda': 49.92817227025565, 'reg__min_split_gain': 2.9270736366858694, 'reg__path_smooth': 3.3524941503404007, 'reg__min_child_weight': 0.3591013914689764, 'reg__feature_fraction': 0.7307688821304147, 'reg__bagging_fraction': 0.5061574428948266, 'reg__bagging_freq': 7, 'reg__max_bin': 80, 'reg__min_data_per_group': 141}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.018739192005699082, 'reg__num_leaves': 41, 'reg__subsample': 0.628720141928181, 'reg__colsample_bytree': 0.7527002666510639, 'reg__min_child_samples': 25, 'reg__reg_alpha': 9.385570265254183, 'reg__reg_lambda': 47.28992980097692, 'reg__min_split_gain': 4.84265674628498, 'reg__path_smooth': 3.2458999937532904, 'reg__min_child_weight': 0.6141105818373385, 'reg__feature_fraction': 0.7812812252437263, 'reg__bagging_fraction': 0.5221649573294369, 'reg__bagging_freq': 4, 'reg__max_bin': 75, 'reg__min_data_per_group': 161}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.04461790007763139, 'reg__num_leaves': 18, 'reg__subsample': 0.8113082931589802, 'reg__colsample_bytree': 0.8124132780725517, 'reg__min_child_samples': 33, 'reg__reg_alpha': 9.453258476563121, 'reg__reg_lambda': 48.456934646268856, 'reg__min_split_gain': 1.9688111707182725, 'reg__path_smooth': 0.002295799747079816, 'reg__min_child_weight':

0.7851798581838736, 'reg__feature_fraction': 0.8171353826748482, 'reg__bagging_fraction': 0.5269196752407124, 'reg__bagging_freq': 6, 'reg__max_bin': 76, 'reg__min_data_per_group': 154}

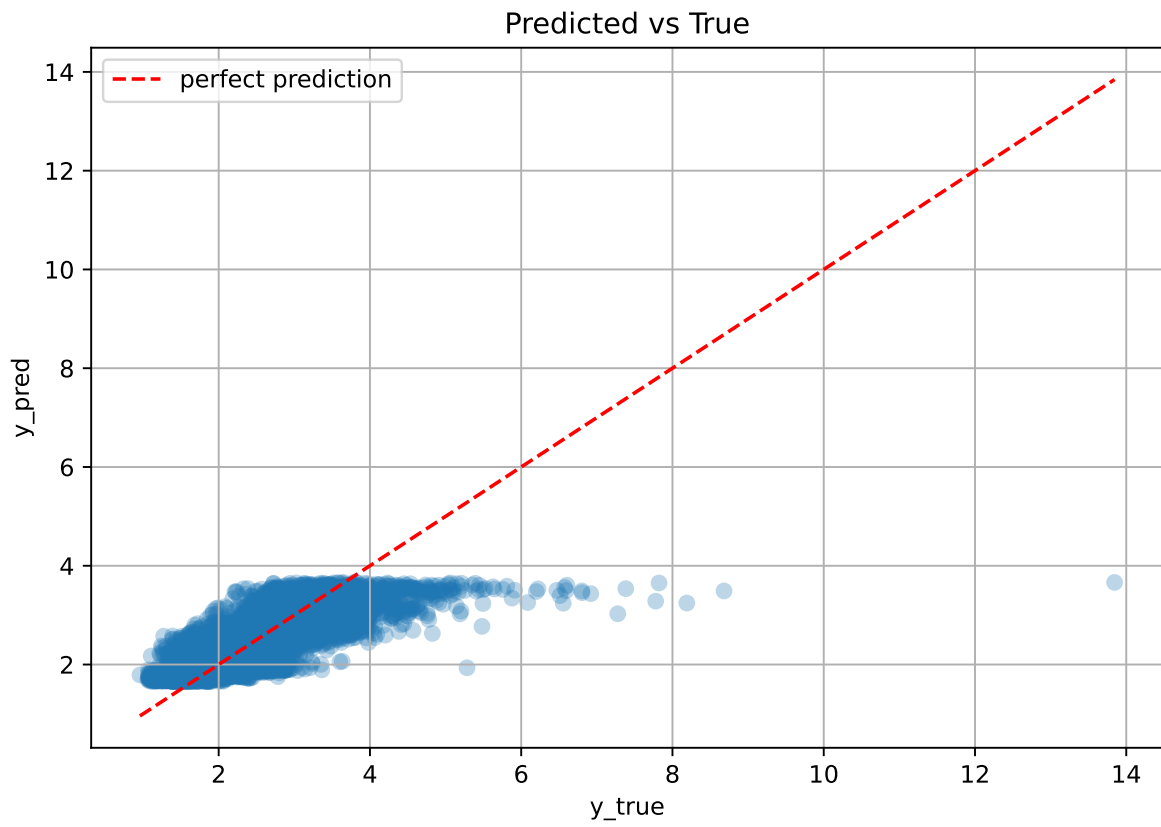
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.04873793648069606, 'reg__num_leaves': 17, 'reg__subsample': 0.8315422141204134, 'reg__colsample_bytree': 0.5387926427984511, 'reg__min_child_samples': 34, 'reg__reg_alpha': 1.9647810713281668, 'reg__reg_lambda': 49.98115903424011, 'reg__min_split_gain': 2.854703936288725, 'reg__path_smooth': 3.3823893801800113, 'reg__min_child_weight': 0.34306652748574196, 'reg__feature_fraction': 0.7348159293266922, 'reg__bagging_fraction': 0.500433389615909, 'reg__bagging_freq': 7, 'reg__max_bin': 79, 'reg__min_data_per_group': 141}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.04266894768560113, 'reg__num_leaves': 17, 'reg__subsample': 0.7999977660653607, 'reg__colsample_bytree': 0.54800970494949, 'reg__min_child_samples': 35, 'reg__reg_alpha': 9.454448420141789, 'reg__reg_lambda': 47.9598251392519, 'reg__min_split_gain': 3.06647850807786, 'reg__path_smooth': 3.1843879246435285, 'reg__min_child_weight': 0.5551284767092333, 'reg__feature_fraction': 0.8104200280675846, 'reg__bagging_fraction': 0.5251726172086038, 'reg__bagging_freq': 5, 'reg__max_bin': 137, 'reg__min_data_per_group': 162}

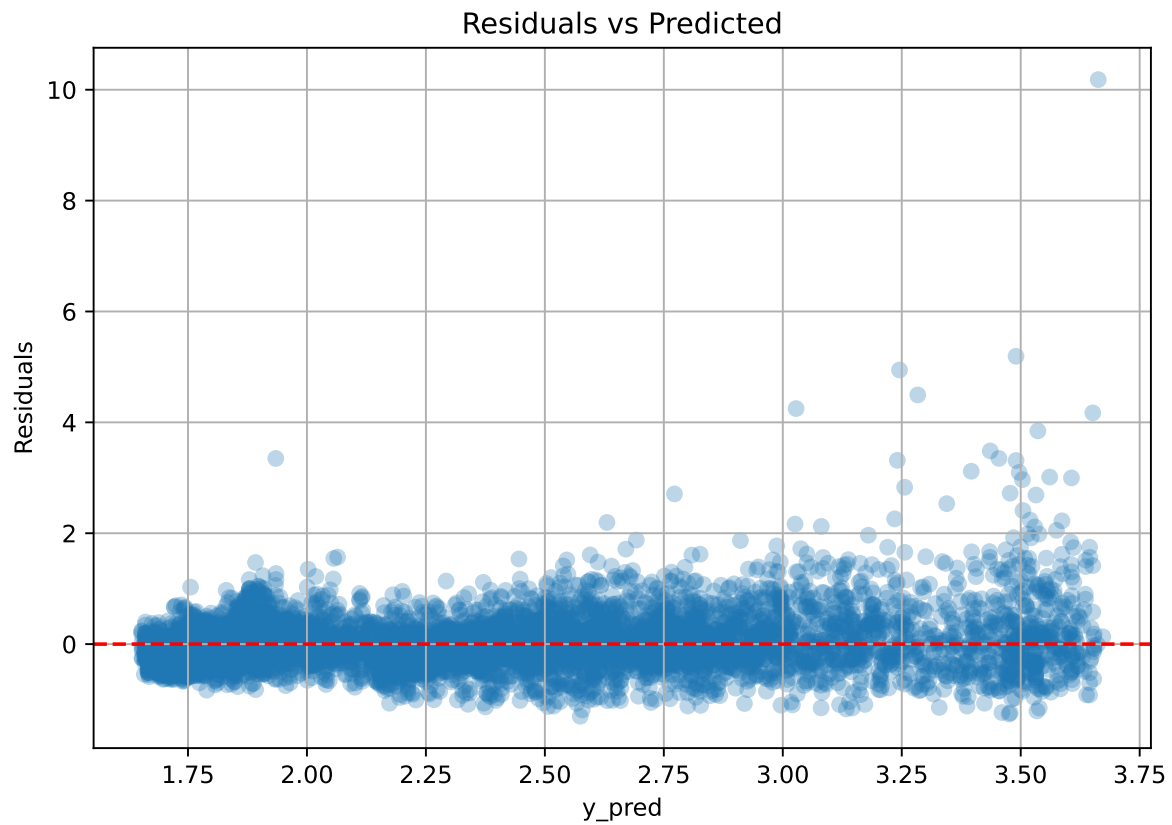
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.018879405992065508, 'reg__num_leaves': 57, 'reg__subsample': 0.6383084271066025, 'reg__colsample_bytree': 0.6574300164951958, 'reg__min_child_samples': 13, 'reg__reg_alpha': 9.935846983736848, 'reg__reg_lambda': 49.43724436016725, 'reg__min_split_gain': 3.6508020123104727, 'reg__path_smooth': 0.05017605391181723, 'reg__min_child_weight': 0.004066870198932103, 'reg__feature_fraction': 0.7475043625082765, 'reg__bagging_fraction': 0.5029857618331792, 'reg__bagging_freq': 4, 'reg__max_bin': 72, 'reg__min_data_per_group': 152}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.04360939658271438, 'reg__num_leaves': 41, 'reg__subsample': 0.635571284075512, 'reg__colsample_bytree': 0.6325635576629017, 'reg__min_child_samples': 33, 'reg__reg_alpha': 9.430832504440227, 'reg__reg_lambda': 48.29155128049666, 'reg__min_split_gain': 2.7092354815738746, 'reg__path_smooth': 2.4524730616331296, 'reg__min_child_weight': 0.6334811267118671, 'reg__feature_fraction': 0.8162192717461657, 'reg__bagging_fraction': 0.5215322517533597, 'reg__bagging_freq': 6, 'reg__max_bin': 134, 'reg__min_data_per_group': 156}

3.5.2 Scatter



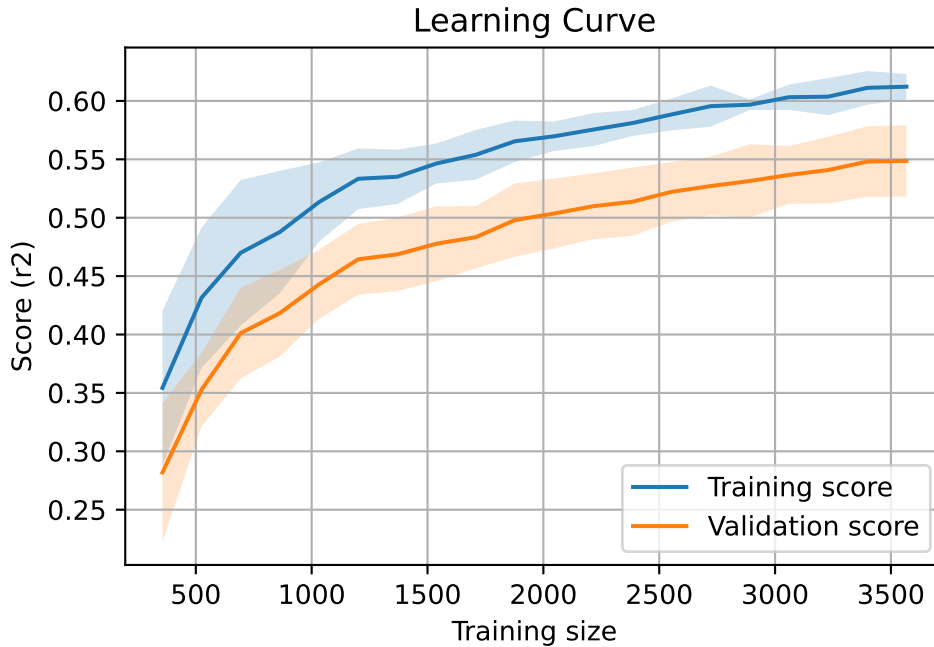
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7280, `Len(training)` : 5096,

`Len(training_real)` : 3568, `Len(test_of_training)` : 1528, `Len(test)` : 2184

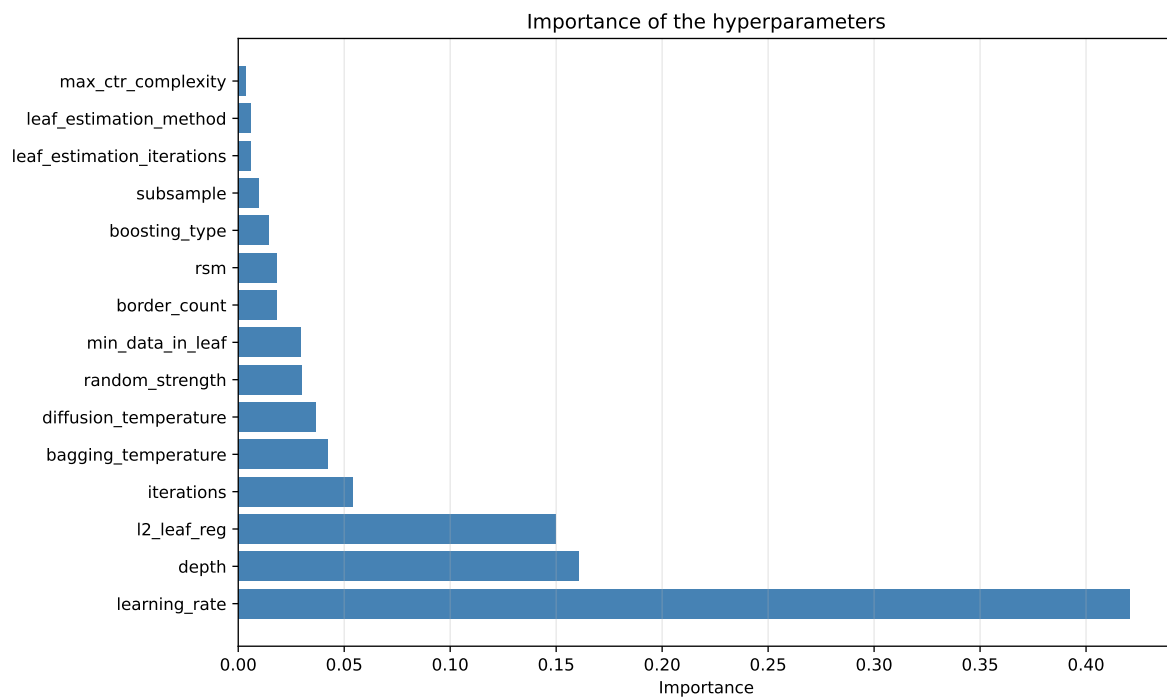
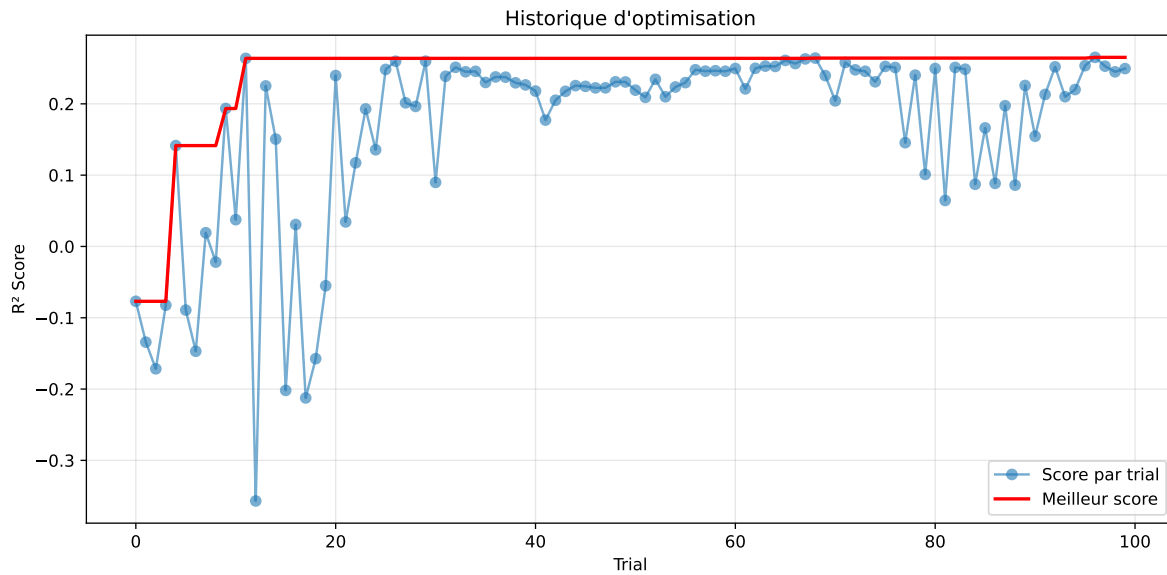


3.6 Model : CatBoost

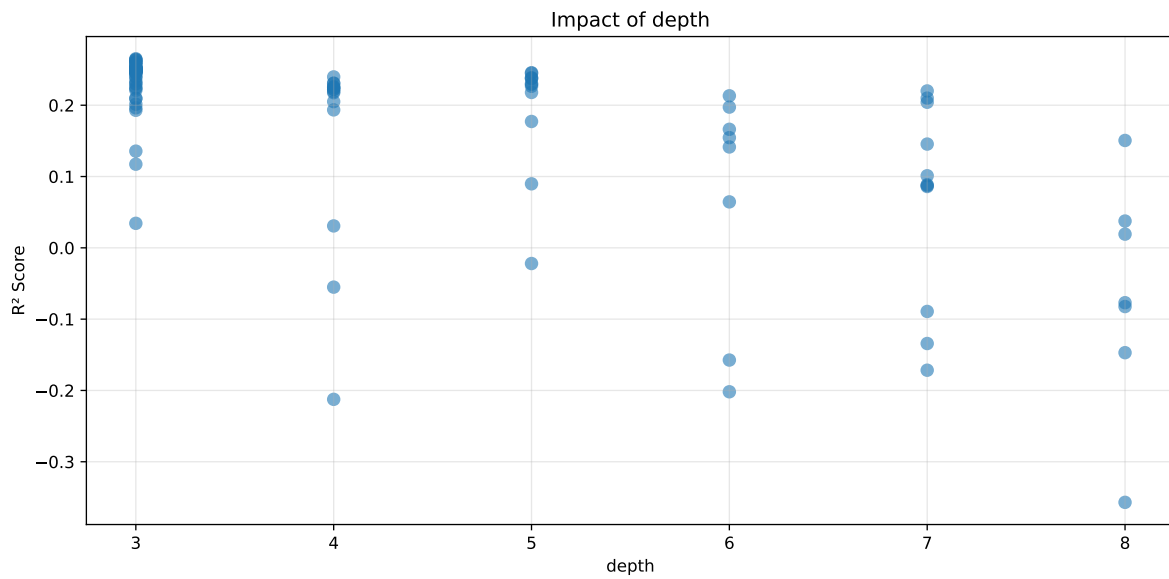
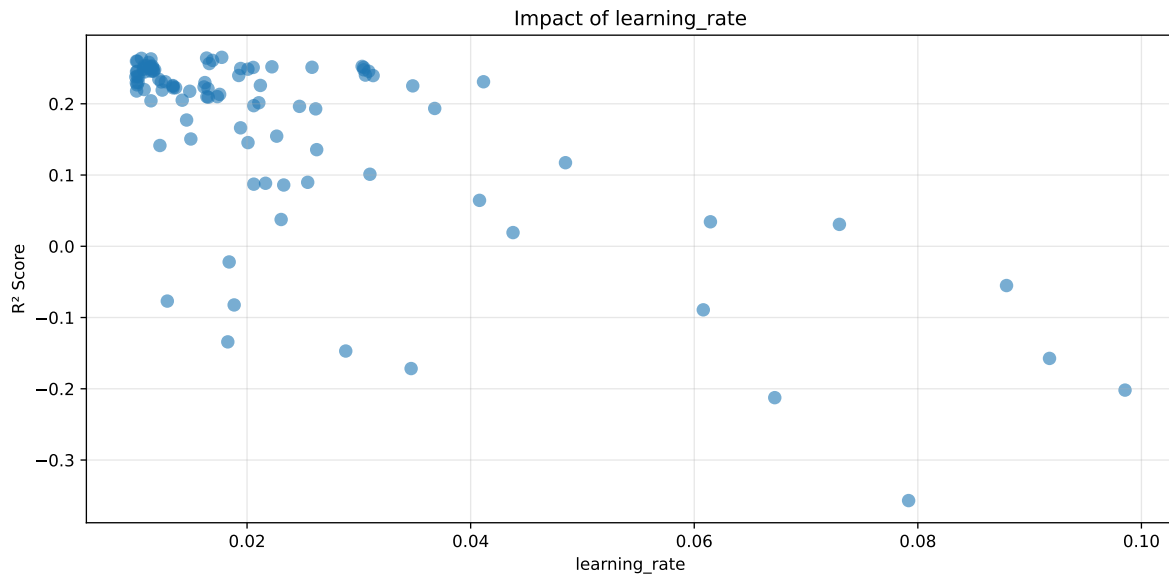
3.6.1 Gridsearch

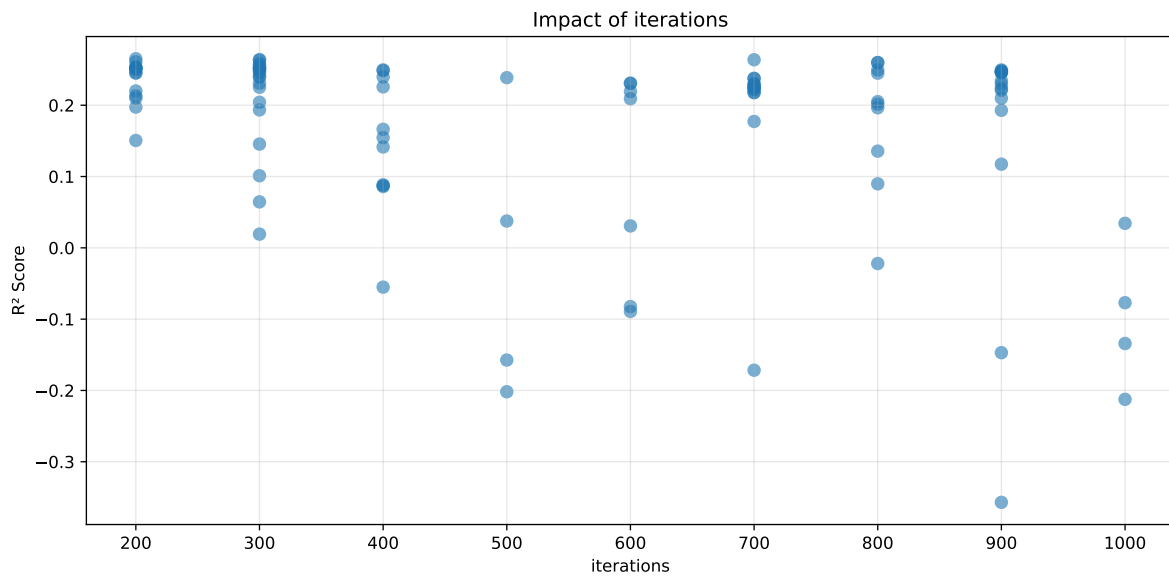
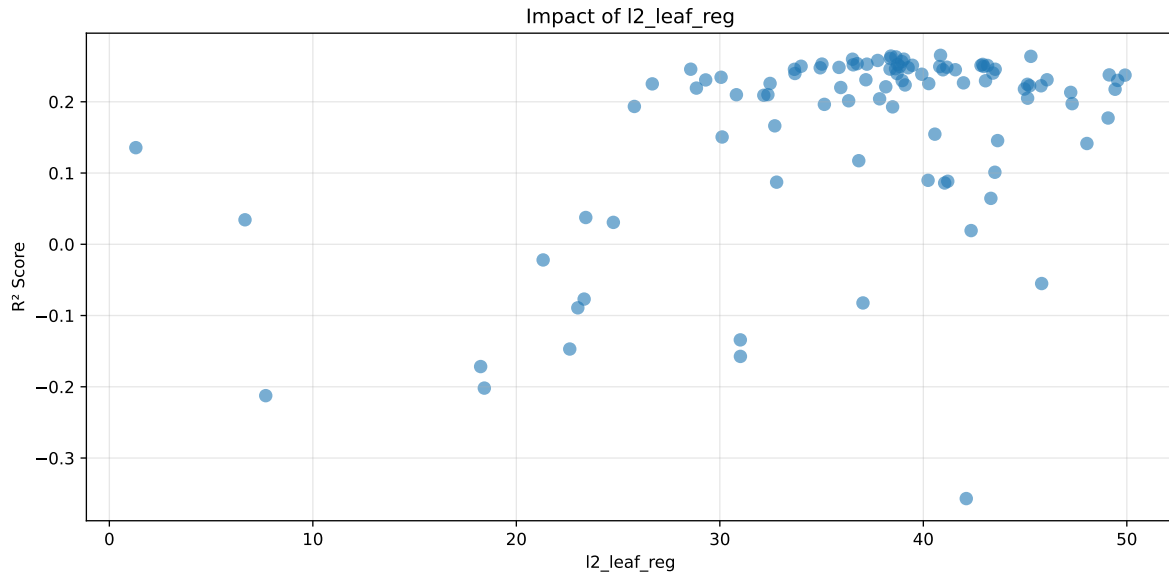
Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845
y_test: mean=2.359, std=0.781, min=1.062, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 28.391618 r_temperature_st_12h_diff_last 20.255411 r_tem-
perature 12.525666 r_temperature_st_8h_slope 9.227205 r_umidity_st_12h_std 6.941703
r_umidity_st_8h_diff_last 5.273599 r_THI 3.271610 r_temperature_st_12h_std
2.717721 r_umidity_st_12h_rms 2.477163 r_NH3 1.608722 r_temperature_st_24h_min
1.584338 r_NH3_st_12h_min 1.043233 r_CO2 0.917198 r_NH3_st_12h_slope 0.786830
r_NH3_st_8h_diff_last 0.683199 r_umidity_st_24h_std 0.672630 r_NH3_st_24h_slope
0.583370 r_NH3_st_8h_std 0.517390 r_temperature_st_12h_min 0.261706 r_tempera-
ture_st_8h_std 0.259686



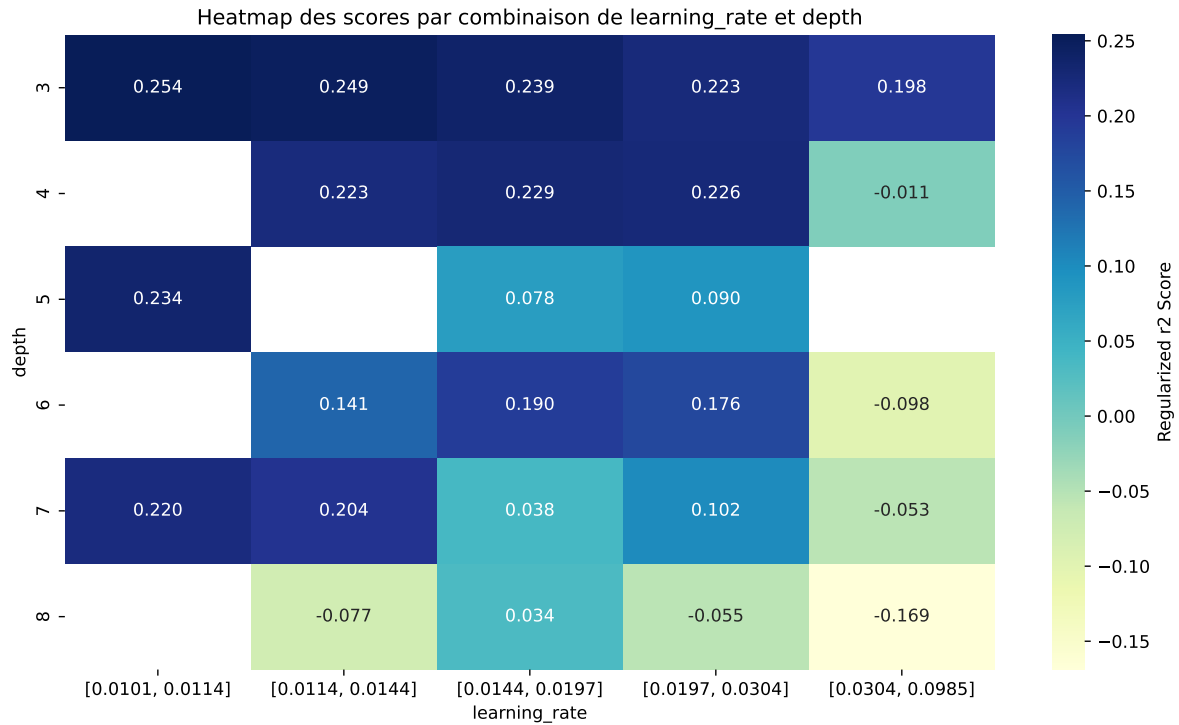
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'l2_leaf_reg', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5915	0.5371	0.6035	0.0299	0.2652	0.6044	0.0674	1.1254	1.1012
2	0.6165	0.5578	0.6248	0.0296	0.2642	0.629	0.0712	1.1277	1.1053
3	0.6281	0.5674	0.6355	0.0293	0.2638	0.6424	0.075	1.1322	1.107
4	0.5872	0.5332	0.6006	0.0298	0.263	0.6048	0.0716	1.1343	1.1014
5	0.5829	0.5292	0.5967	0.0296	0.2609	0.6016	0.0724	1.1368	1.1014
6	0.6444	0.5803	0.648	0.0287	0.2598	0.6543	0.074	1.1276	1.1105
7	0.6423	0.5785	0.6457	0.0294	0.2597	0.6497	0.0712	1.1231	1.1102
8	0.5833	0.5291	0.5967	0.0304	0.258	0.6019	0.0728	1.1377	1.1025
9	0.6193	0.5588	0.627	0.03	0.2565	0.633	0.0742	1.1328	1.1082
10	0.5406	0.4927	0.5575	0.0271	0.2535	0.5635	0.0707	1.1435	1.0971
---	---	---	---	---	---	---	---	---	---
100	0.9944	0.7691	0.8379	0.0305	-0.3569	0.8439	0.0747	1.0971	1.2928

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017738814717595728, 'reg__l2_leaf_reg': 40.84566518777396, 'reg__bagging_temperature': 2.545355964602097, 'reg__random_strength': 3.8293436378227828, 'reg__subsample': 0.6306965800032595, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7145646474579748, 'reg__border_count': 106, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4251.350862368284}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.016381645291227702, 'reg__l2_leaf_reg': 38.407199593428246, 'reg__bagging_temperature': 1.5956830550233296, 'reg__random_strength': 4.858731750091236, 'reg__subsample': 0.5580195795011891, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8224699803058332, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5321.509916031491}

Rank 3: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01055050281928244, 'reg__l2_leaf_reg': 45.28436623564218, 'reg__bagging_temperature': 0.6609944058117351, 'reg__random_strength': 4.490908062128136, 'reg__subsample': 0.5648302159326746, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.665009974631629, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1070.2811159149471}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011403513614925892, 'reg__l2_leaf_reg': 38.65303661401663, 'reg__bagging_temperature': 3.7251372381652788, 'reg__random_strength': 4.428163995084379, 'reg__subsample': 0.5539966976935052, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7010209807713645, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5406.892352139426}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01691570578312715, 'reg__l2_leaf_reg': 38.37423648061854, 'reg__bagging_temperature': 1.422983375002209, 'reg__random_strength': 4.419072123243236, 'reg__subsample': 0.5499429990007517, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.701299832760462, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4790.258368600043}

Rank 6: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.01010728005312343, 'reg__l2_leaf_reg': 39.04103200561126, 'reg__bagging_temperature': 3.6401504743250097, 'reg__random_strength': 3.8736027658338728, 'reg__subsample': 0.5877611469070454, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7549553849000881, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4907.396145340174}

Rank 7: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.010186663683535558, 'reg__l2_leaf_reg': 36.52063998508862, 'reg__bagging_temperature': 3.165871688986199, 'reg__random_strength': 4.015772435600938, 'reg__subsample': 0.8811758896291003,

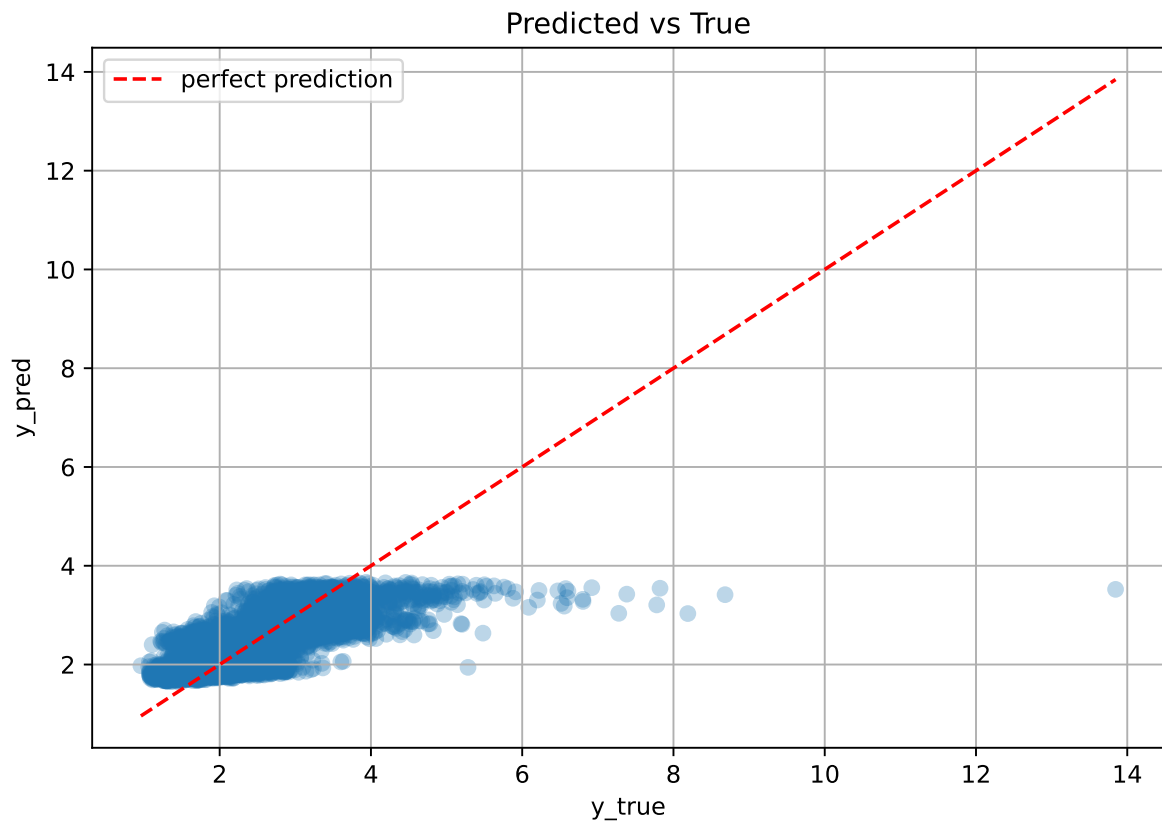
'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6402022013881034, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4495.858548244737}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011216800151462517, 'reg__l2_leaf_reg': 37.757038776837966, 'reg__bagging_temperature': 1.5580539325201082, 'reg__random_strength': 4.762335341755163, 'reg__subsample': 0.5442689301675115, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6987808609514419, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5363.142872886572}

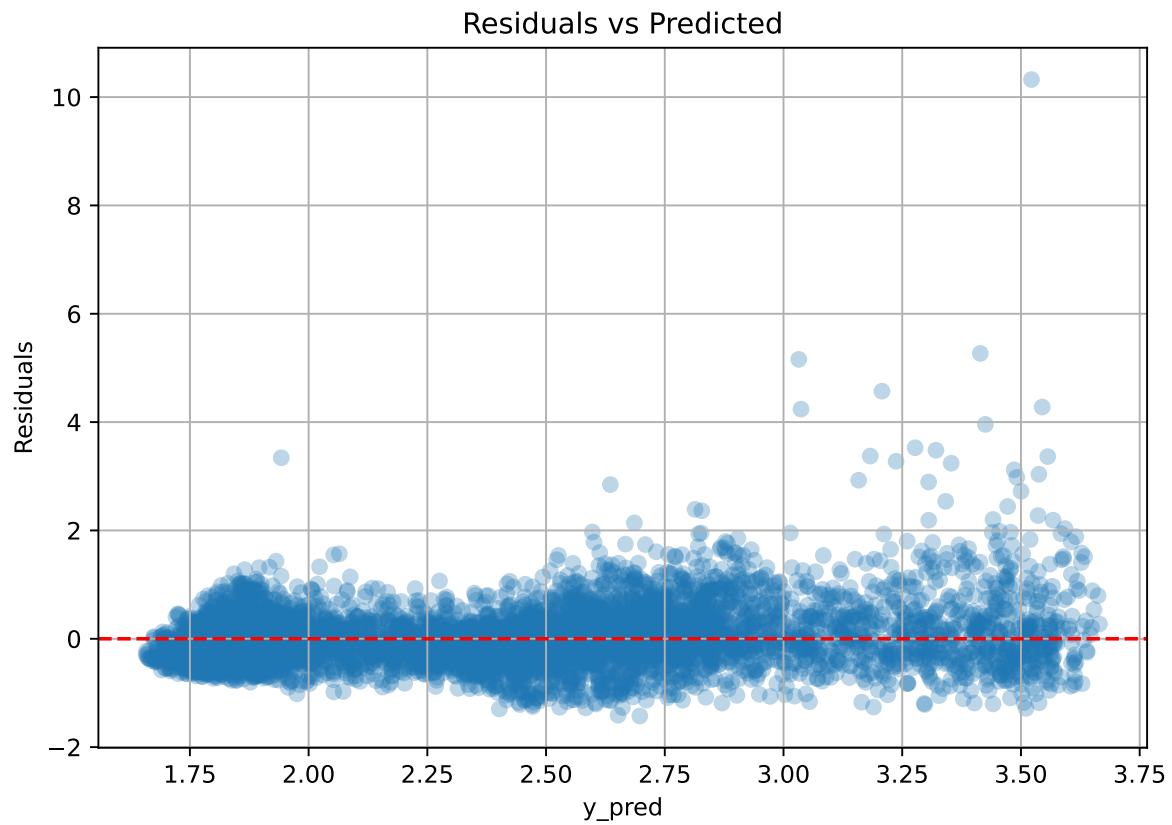
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01661999340672472, 'reg__l2_leaf_reg': 38.94231977864663, 'reg__bagging_temperature': 3.8898857123208317, 'reg__random_strength': 4.416363313161545, 'reg__subsample': 0.5604090252582048, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6934294271728656, 'reg__border_count': 115, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5379.424773177839}

Rank 10: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01102511419120579, 'reg__l2_leaf_reg': 36.74823001440917, 'reg__bagging_temperature': 4.506138268787192, 'reg__random_strength': 4.2228189320697265, 'reg__subsample': 0.795006236442177, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7102576618674042, 'reg__border_count': 107, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4298.851362597254}

3.6.2 Scatter



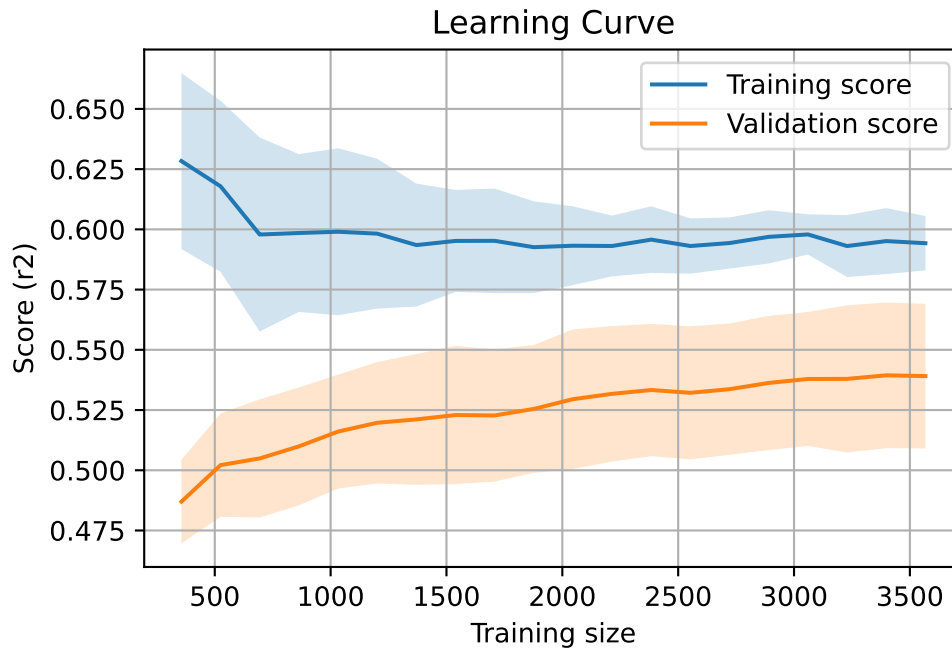
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7280, `Len(training)` : 5096,

`Len(training_real)` : 3568, `Len(test_of_training)` : 1528, `Len(test)` : 2184



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.56	0.505	0.028	0.565	0.566	0.06	1.12	1.109	nan
LGBM	0.614	0.552	0.033	0.624	0.648	0.096	1.174	1.112	0.2423
CatBoost	0.592	0.537	0.03	0.604	0.604	0.067	1.125	1.101	0.2652

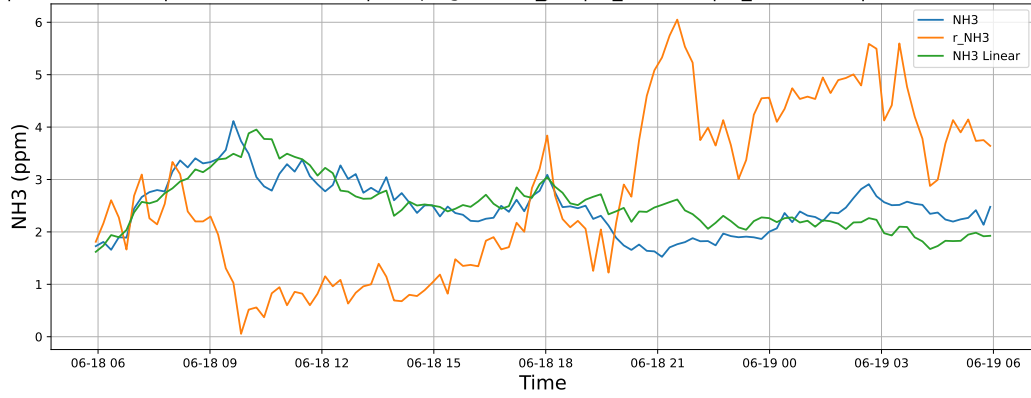
4 Plot

4.1 Standalone plots

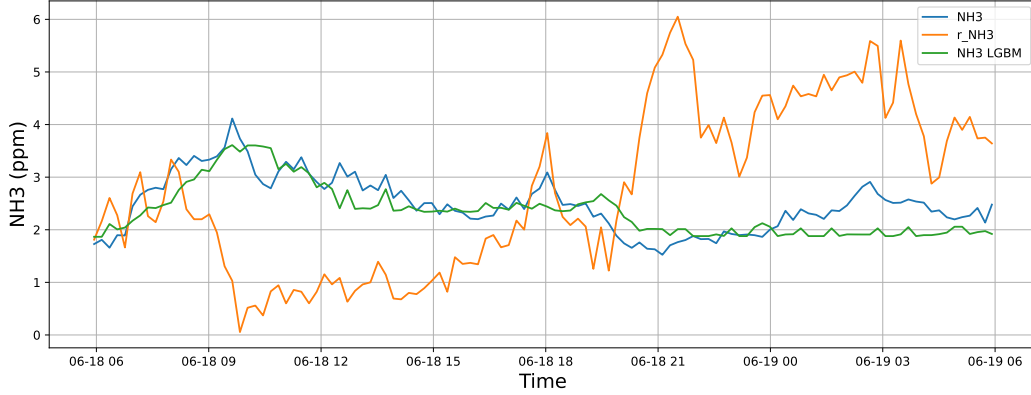
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

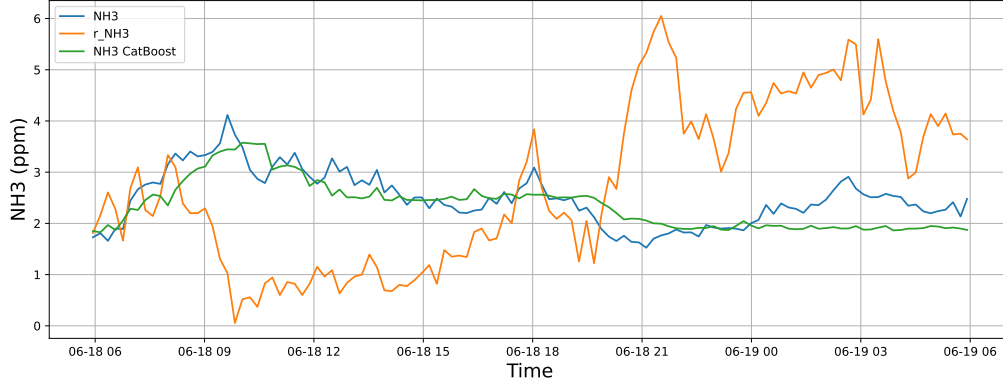
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

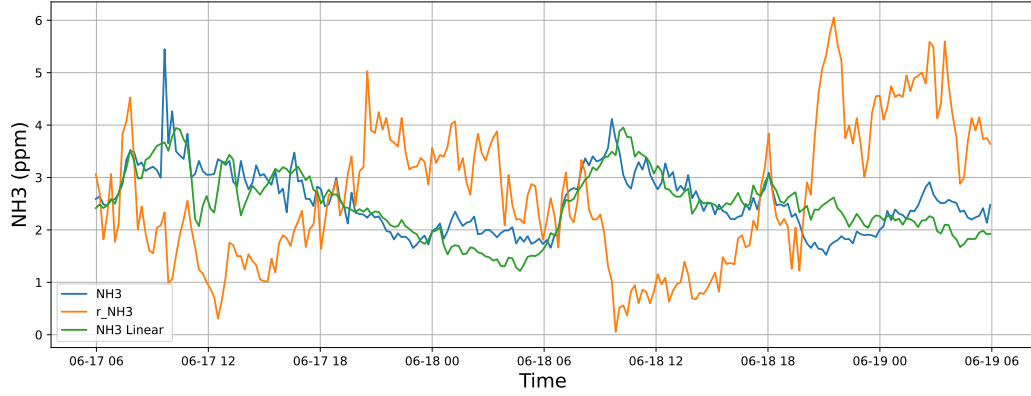


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

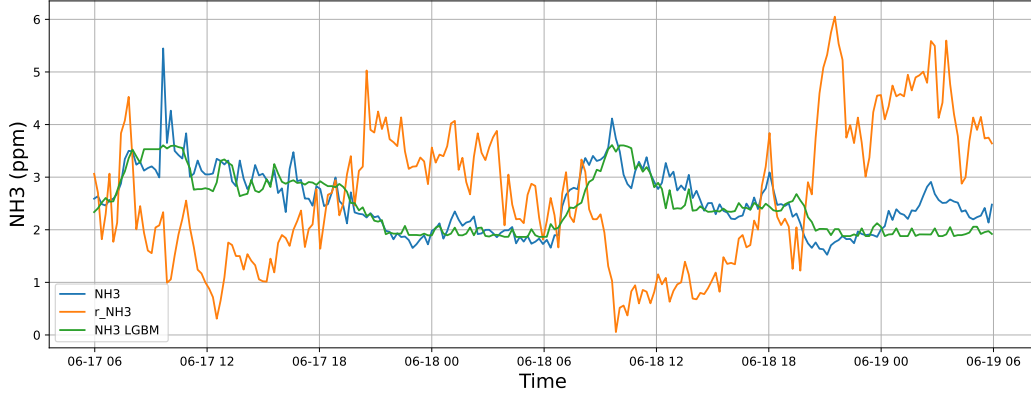


4.1.1.2 Time window : 48

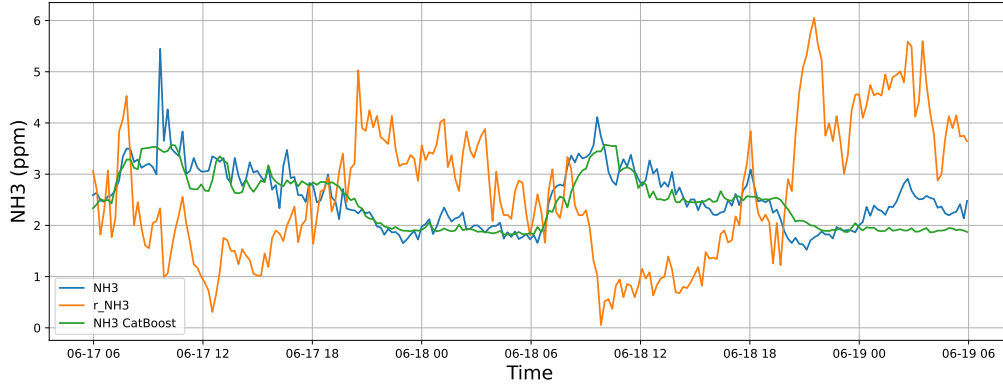
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

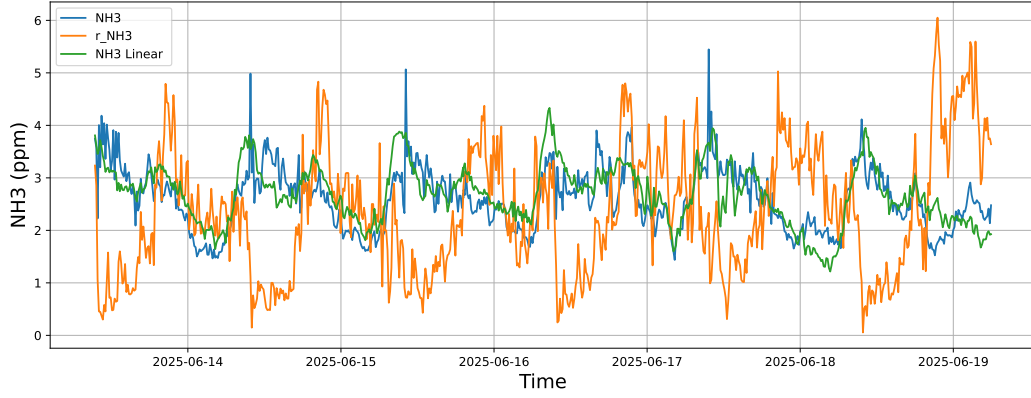


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

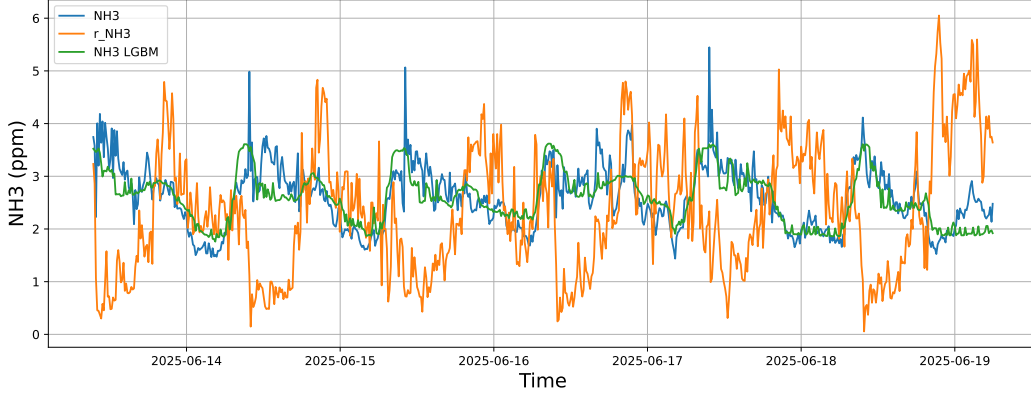


4.1.1.3 Time window : ALL

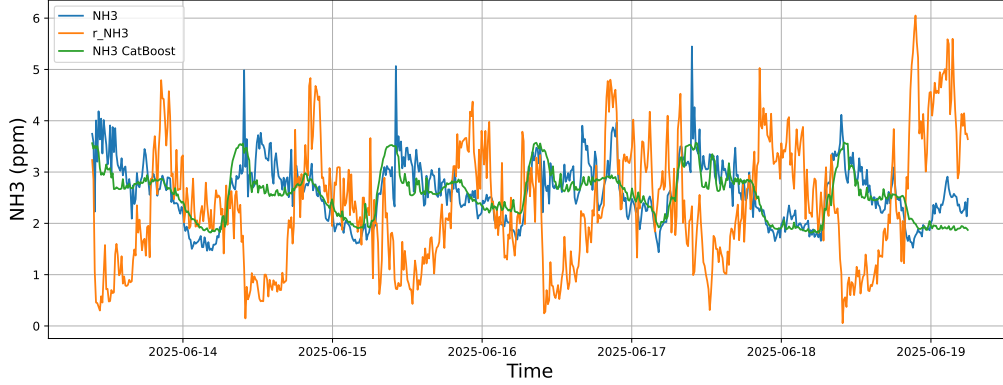
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



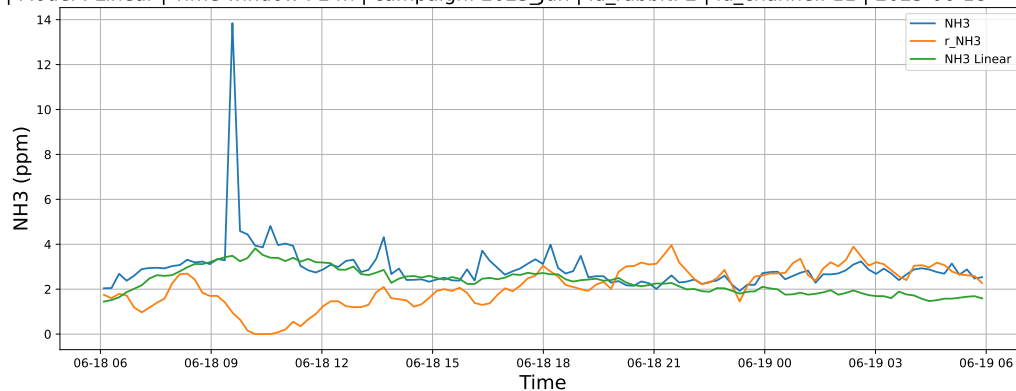
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



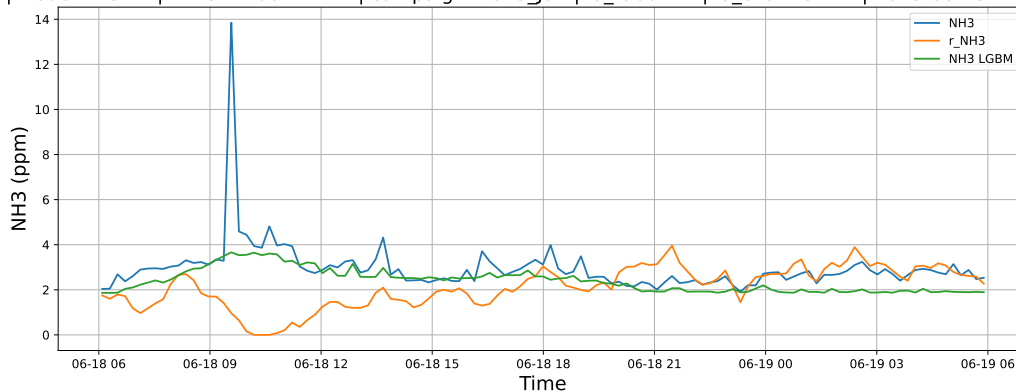
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

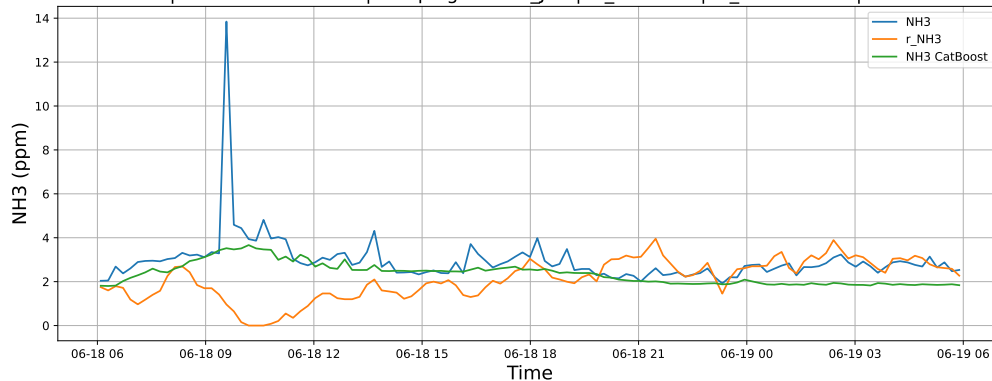
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

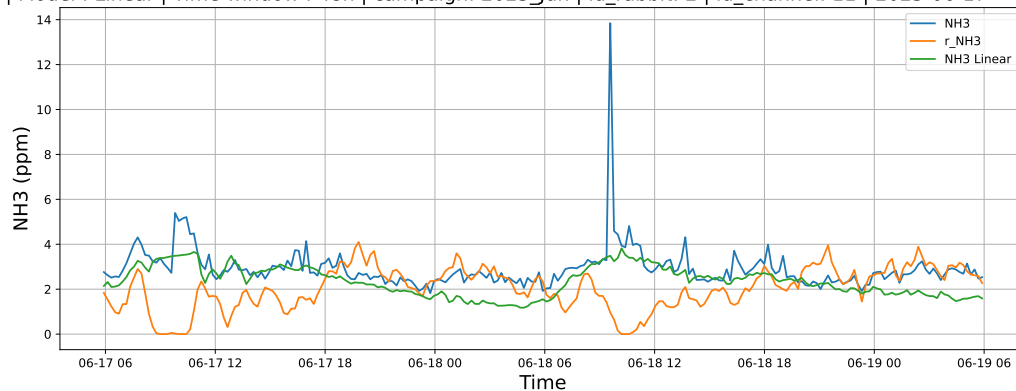


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

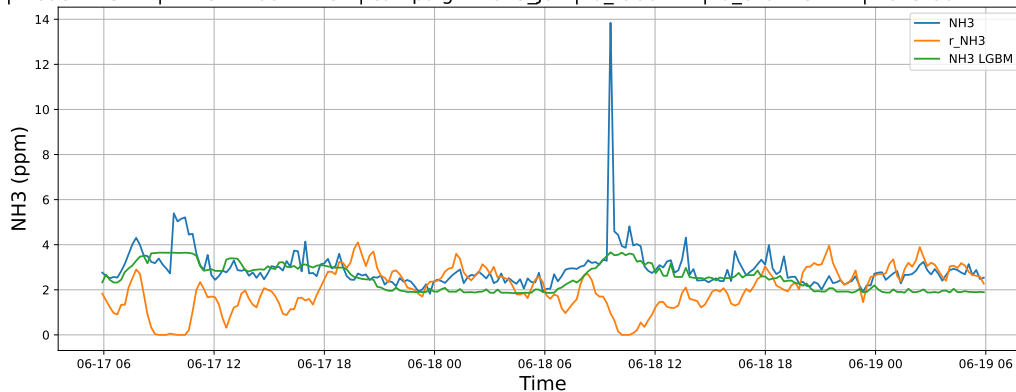


4.1.2.2 Time window : 48

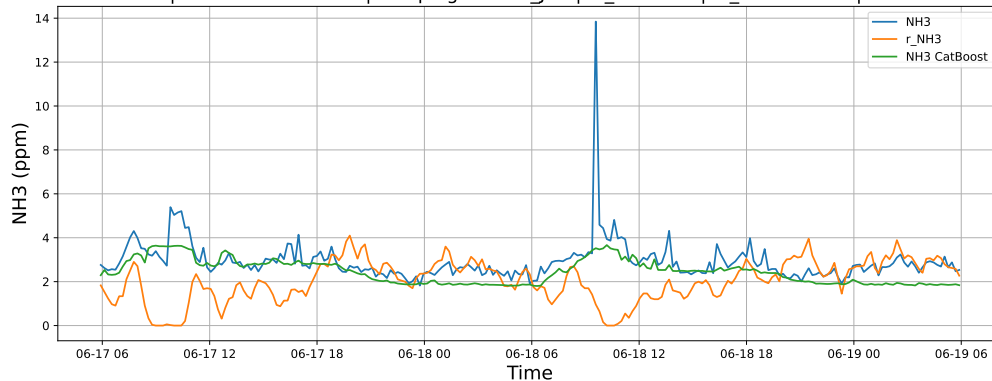
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

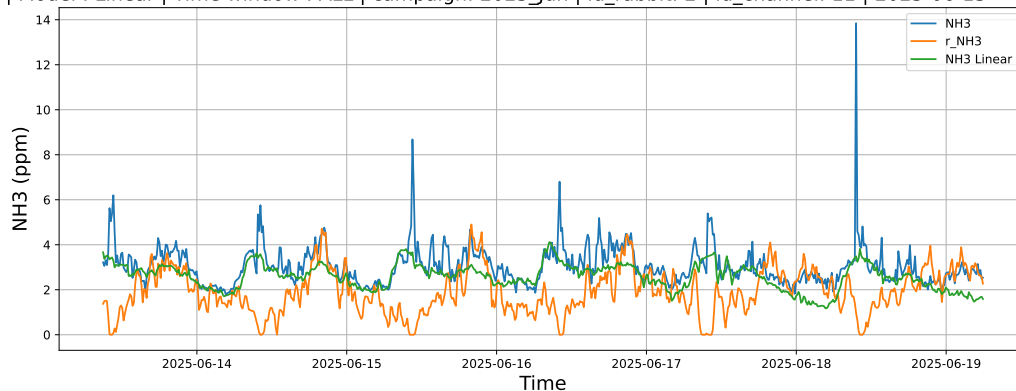


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

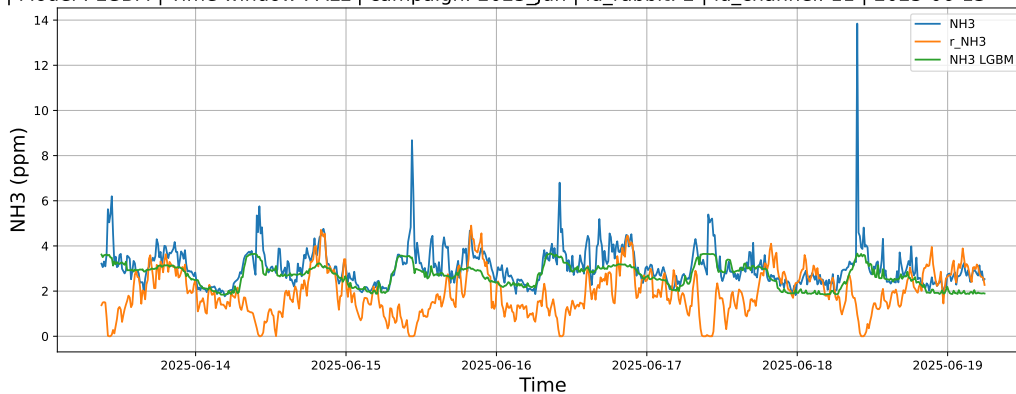


4.1.2.3 Time window : ALL

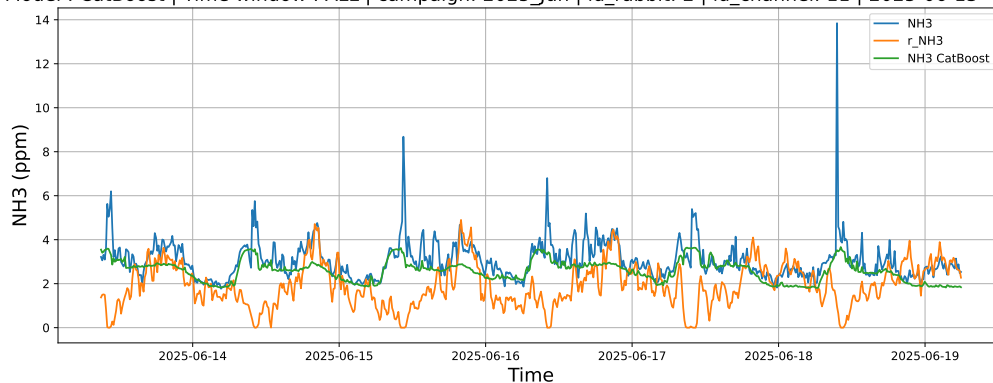
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



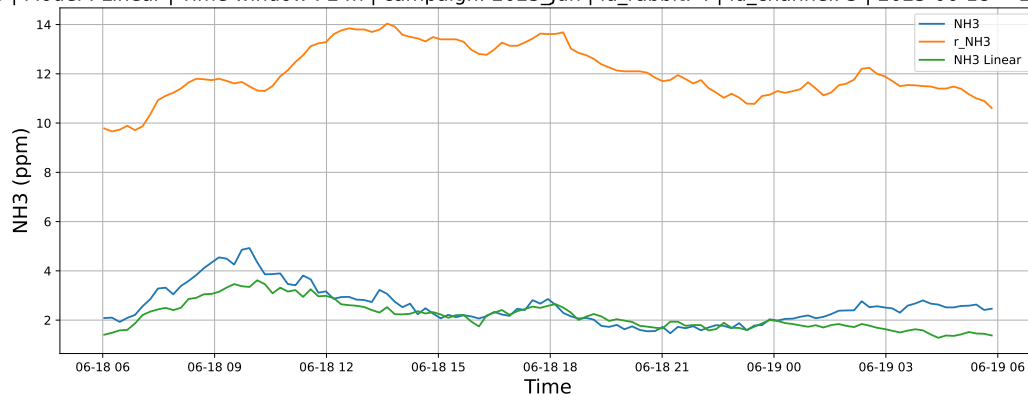
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



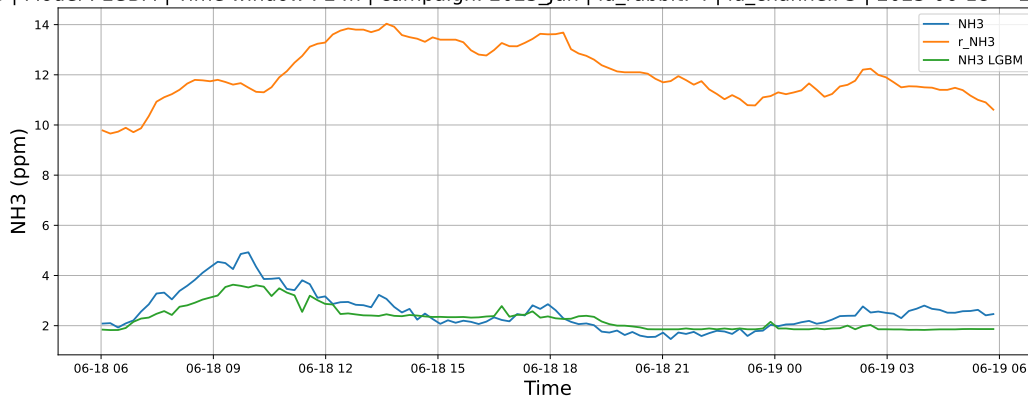
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

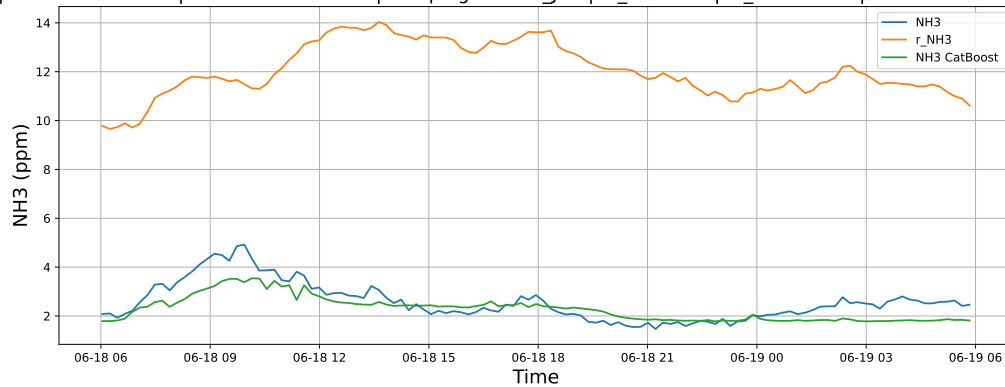
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

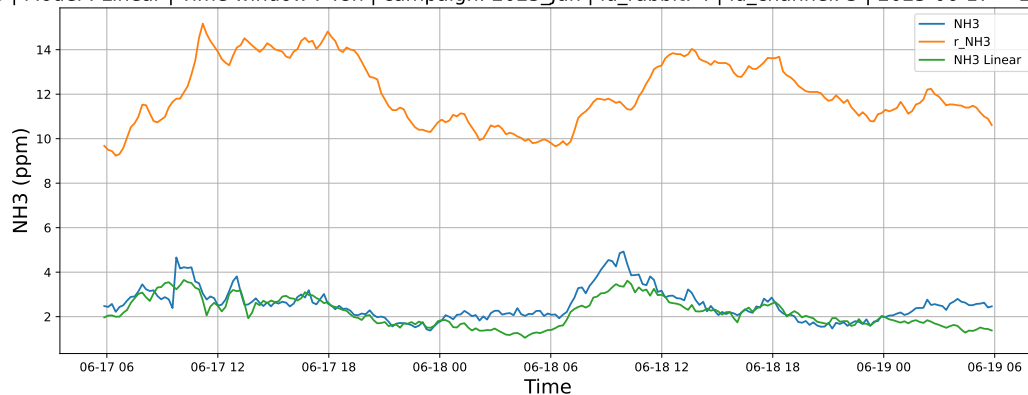


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

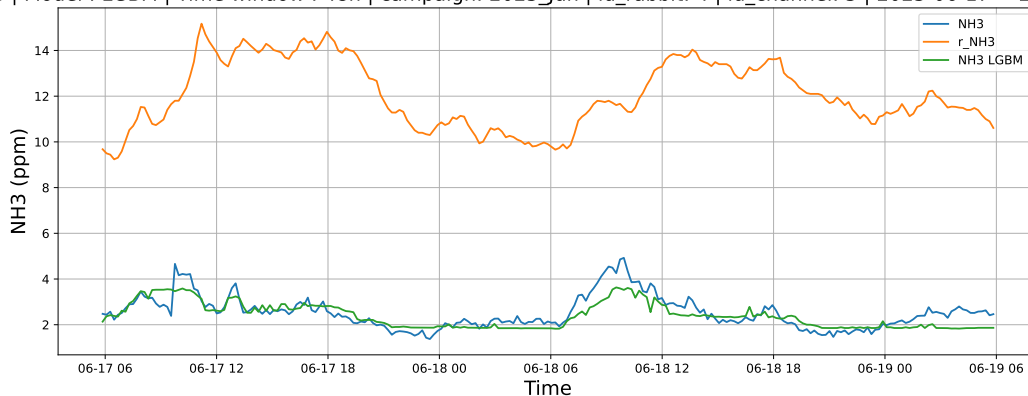


4.1.3.2 Time window : 48

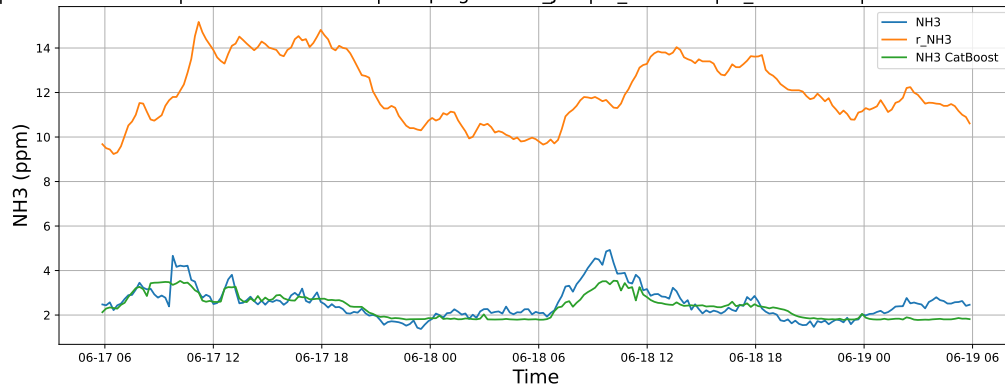
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

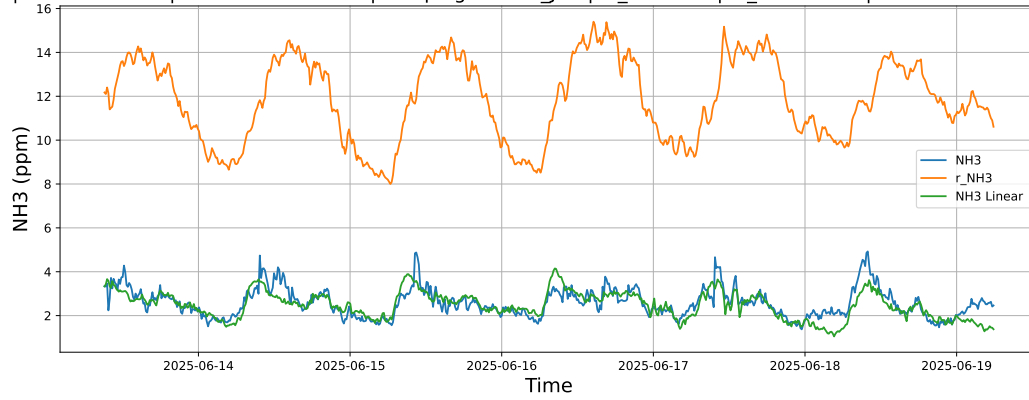


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

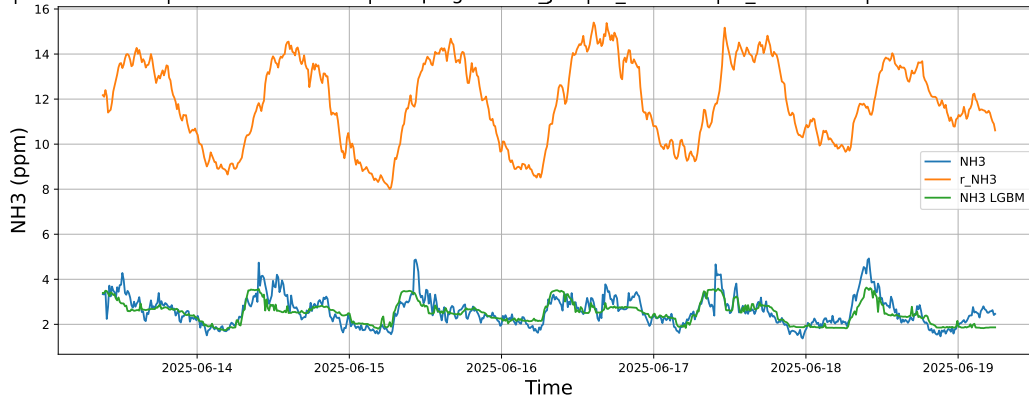


4.1.3.3 Time window : ALL

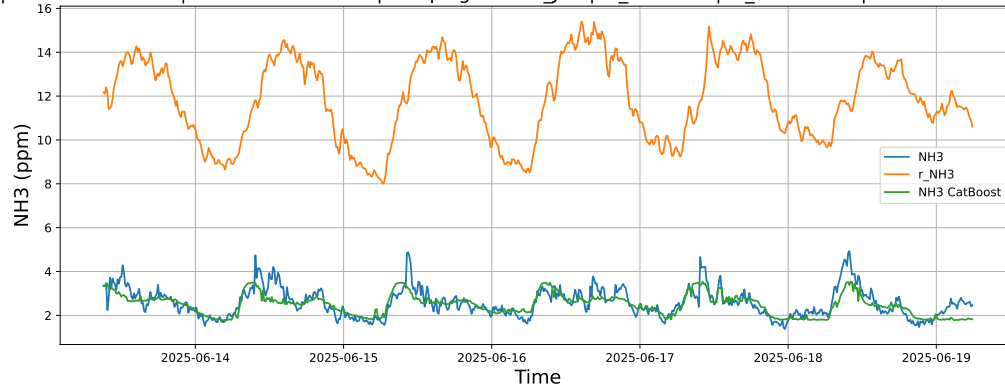
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



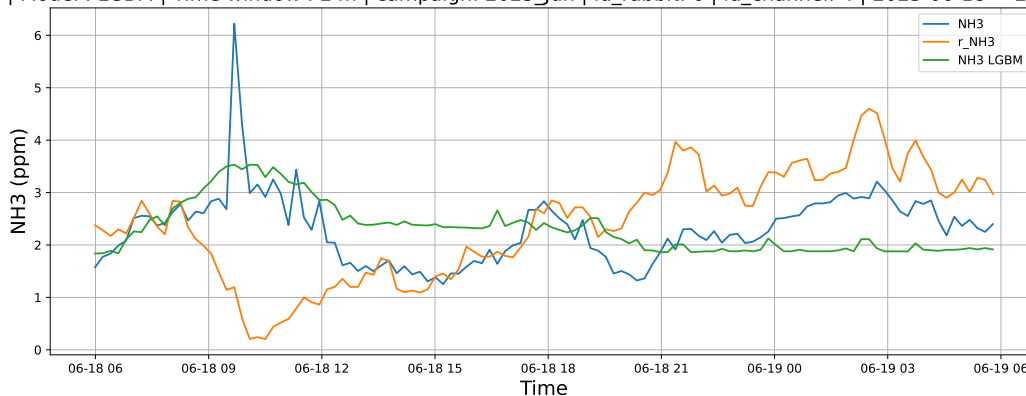
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

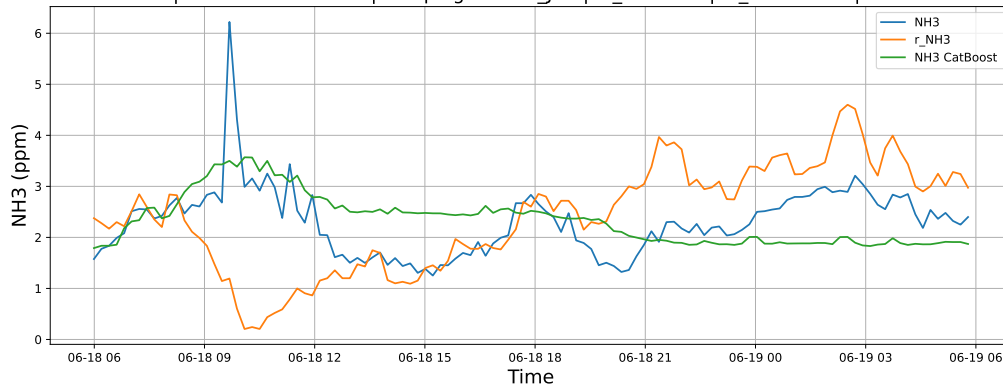
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

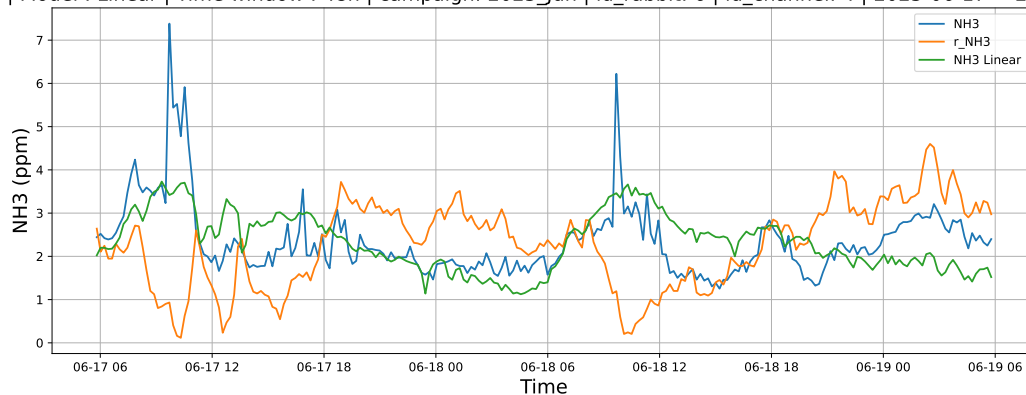


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

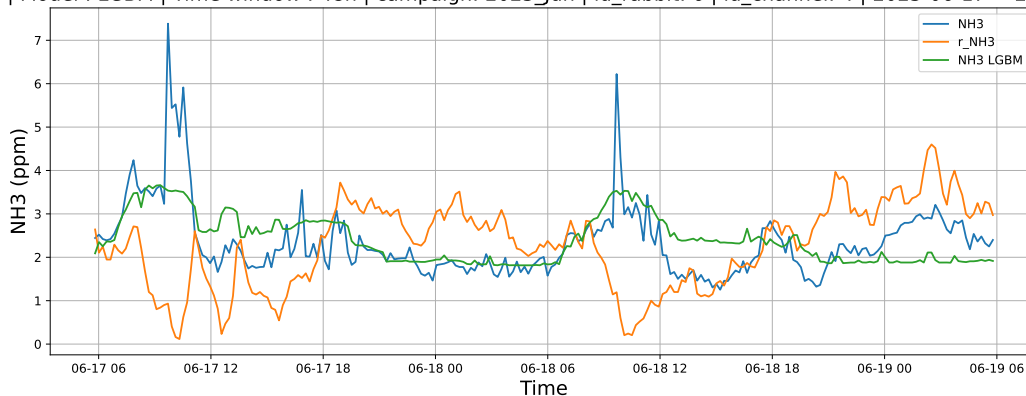


4.1.4.2 Time window : 48

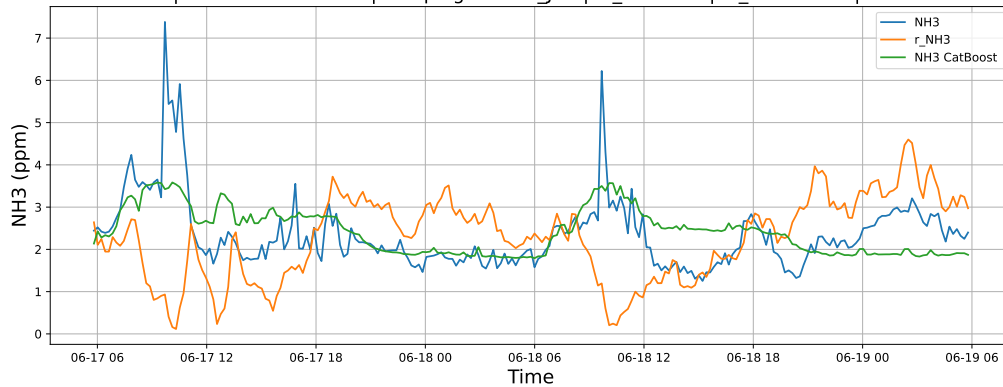
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

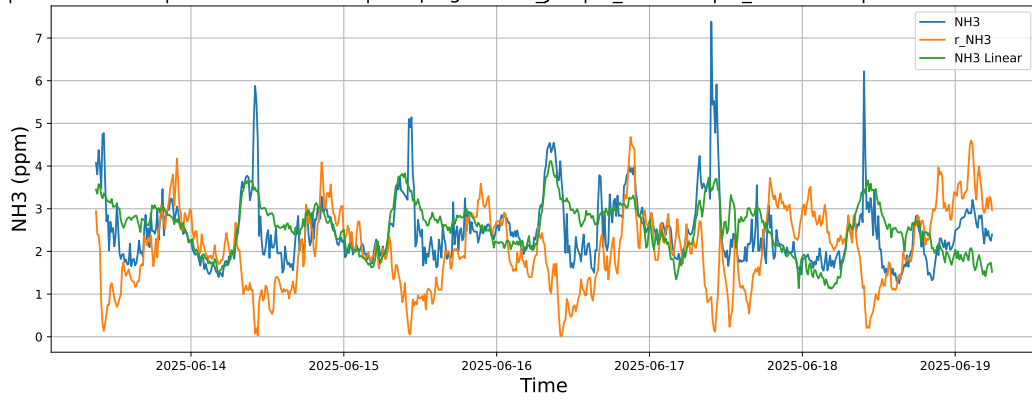


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

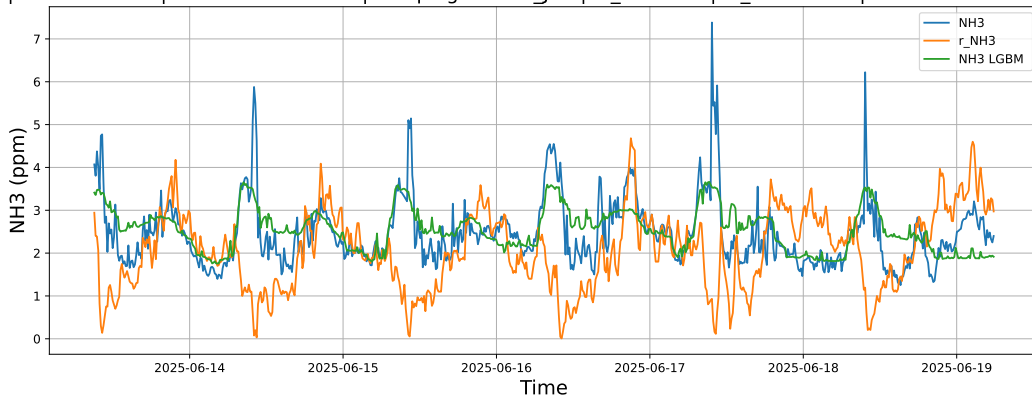


4.1.4.3 Time window : ALL

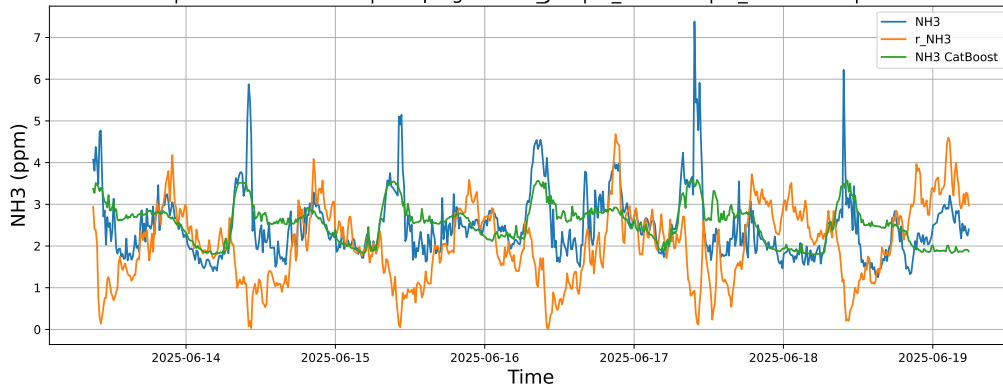
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



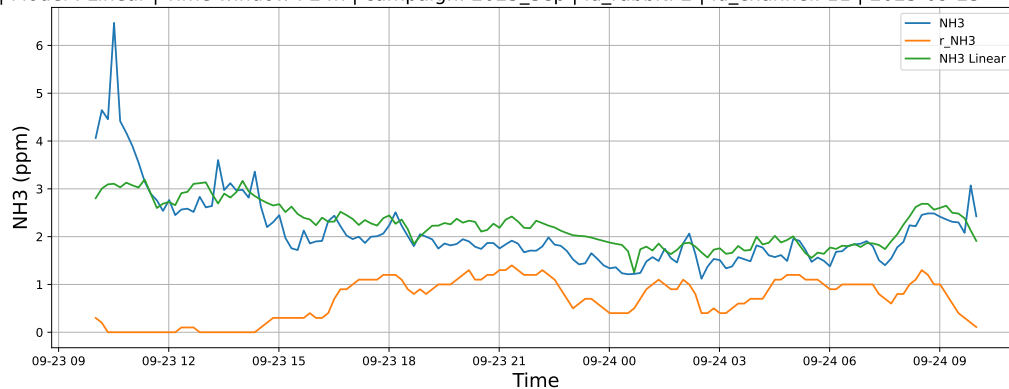
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



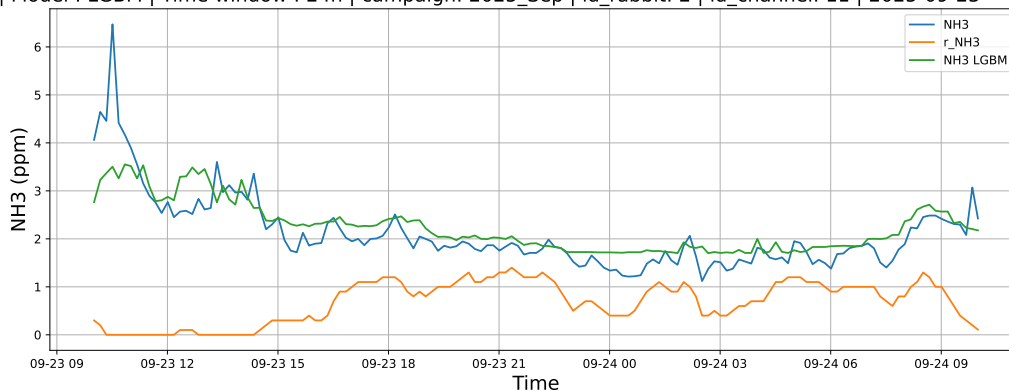
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

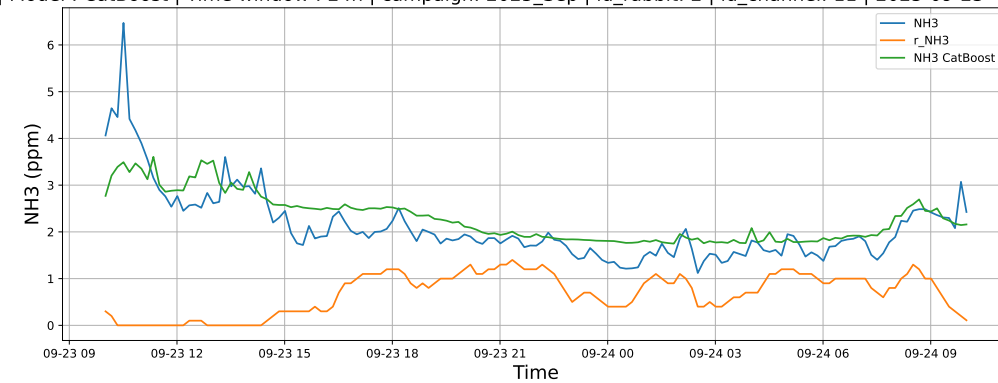
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

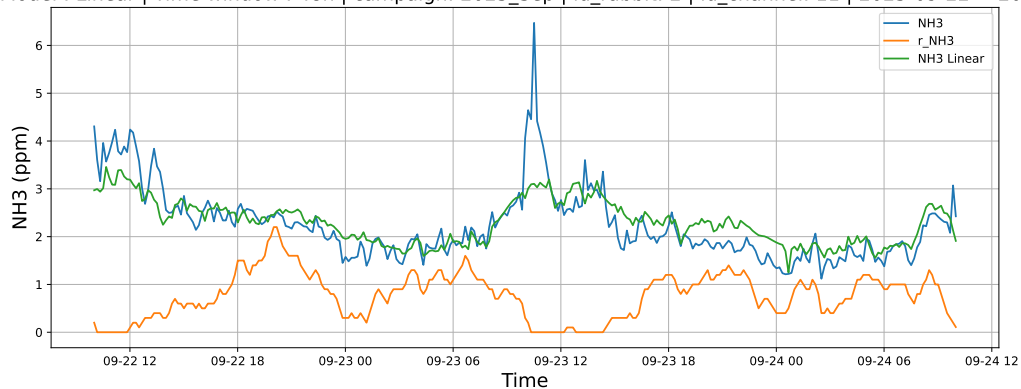


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

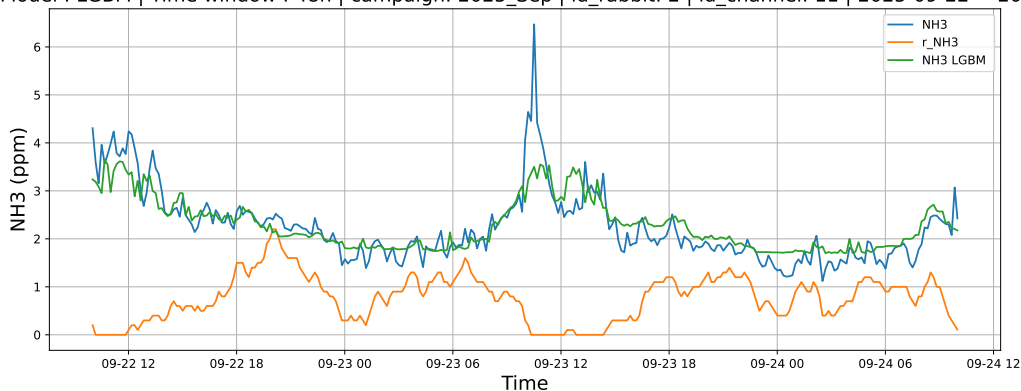


4.1.5.2 Time window : 48

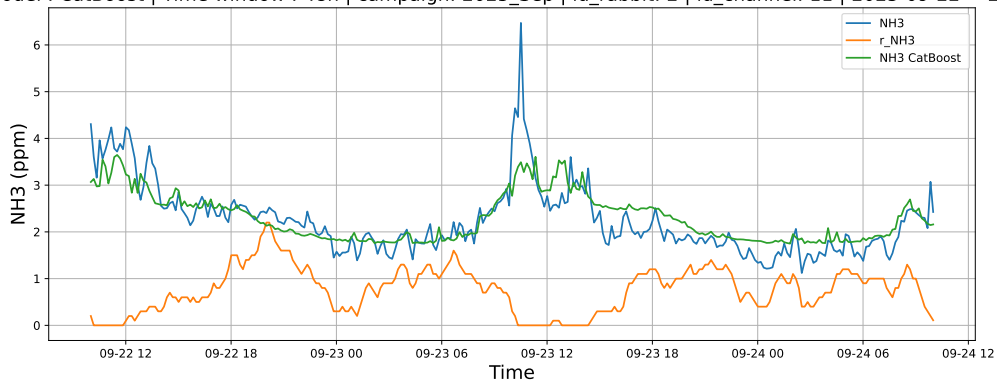
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

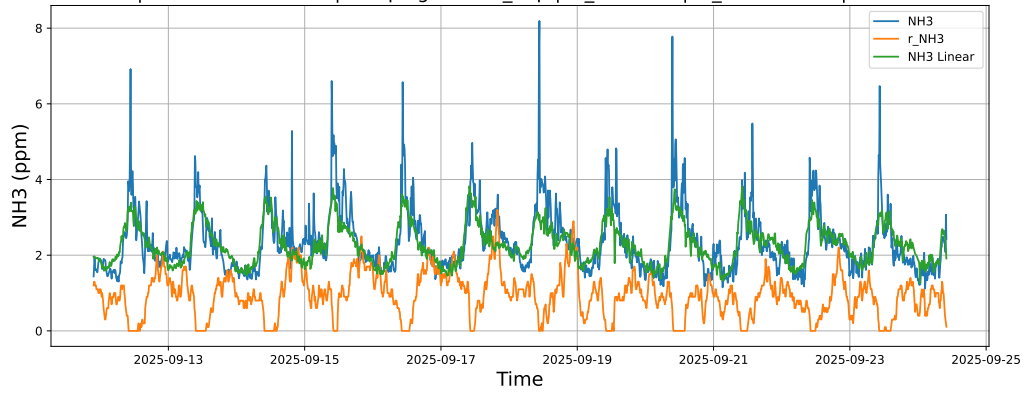


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

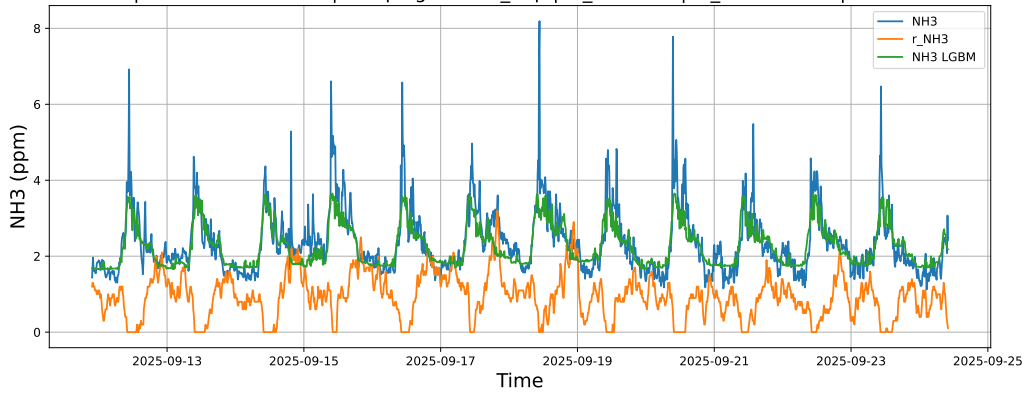


4.1.5.3 Time window : ALL

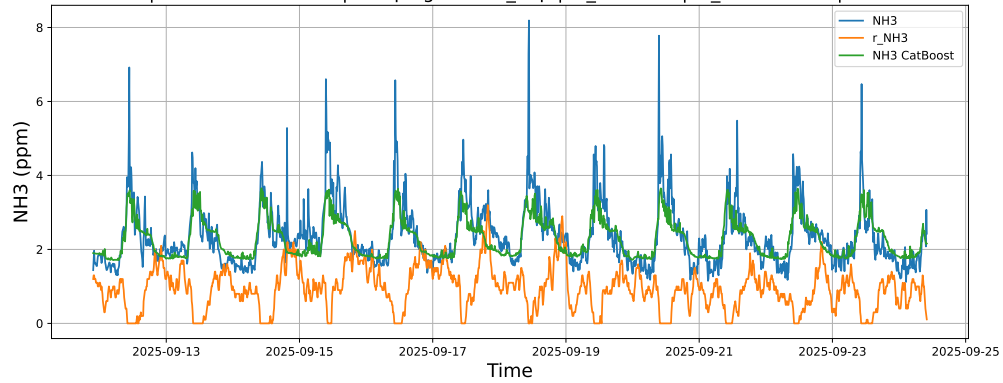
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



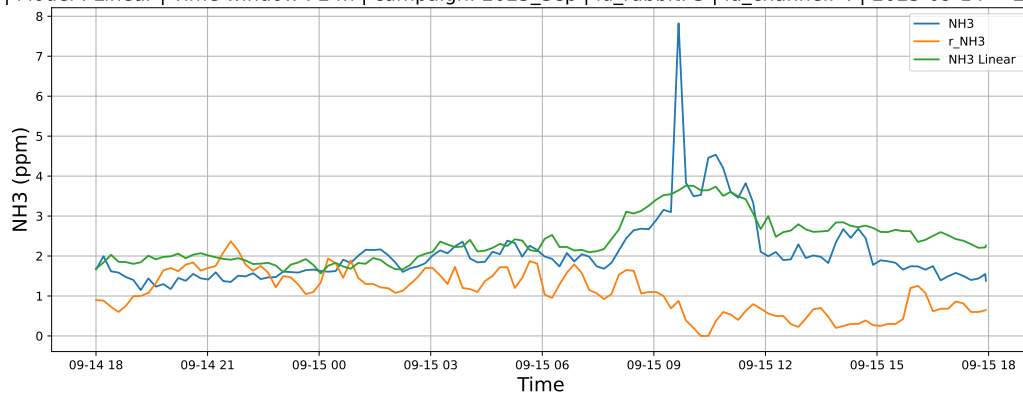
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



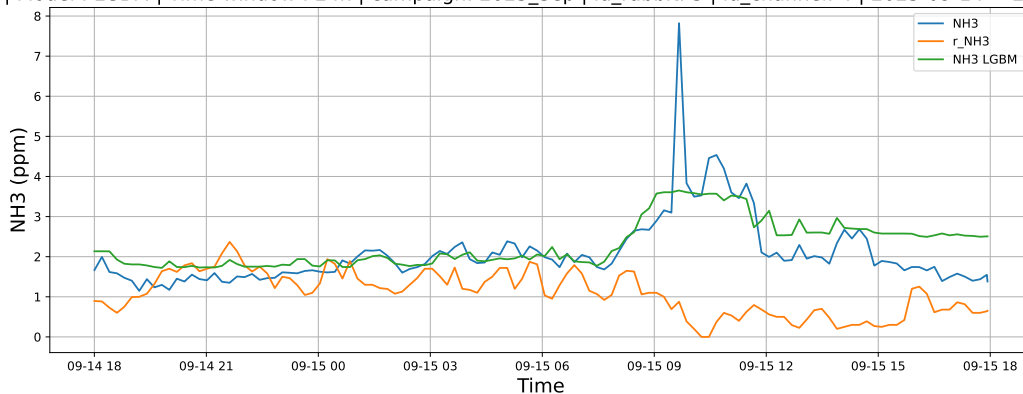
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

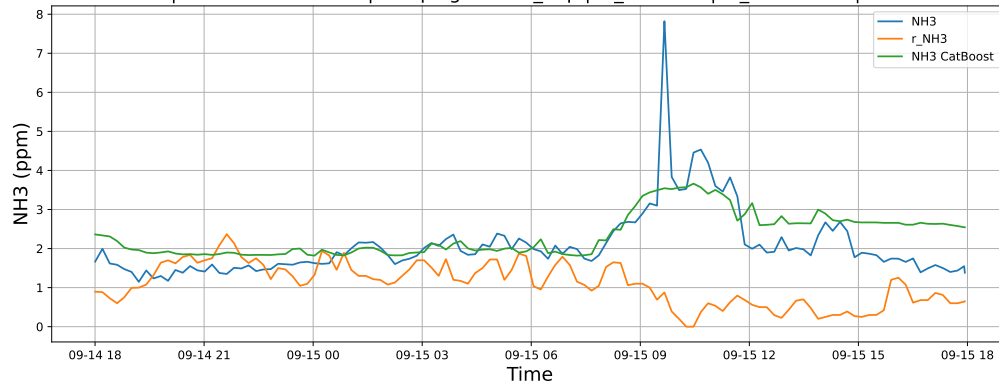
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

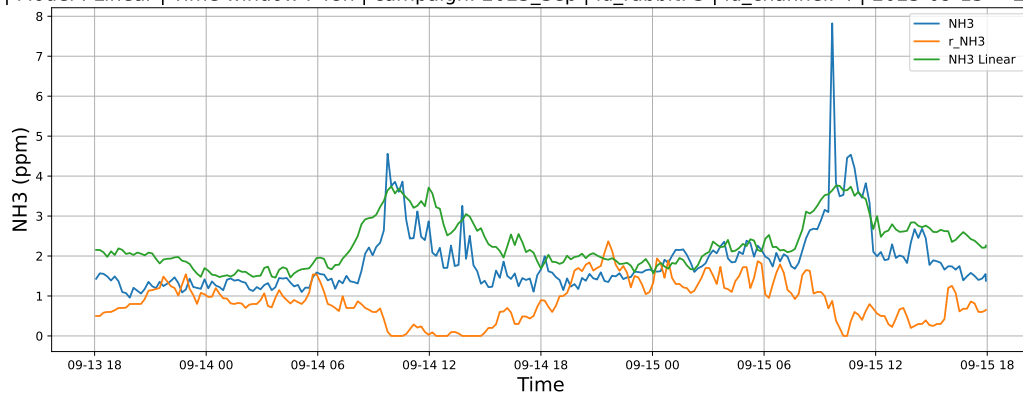


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

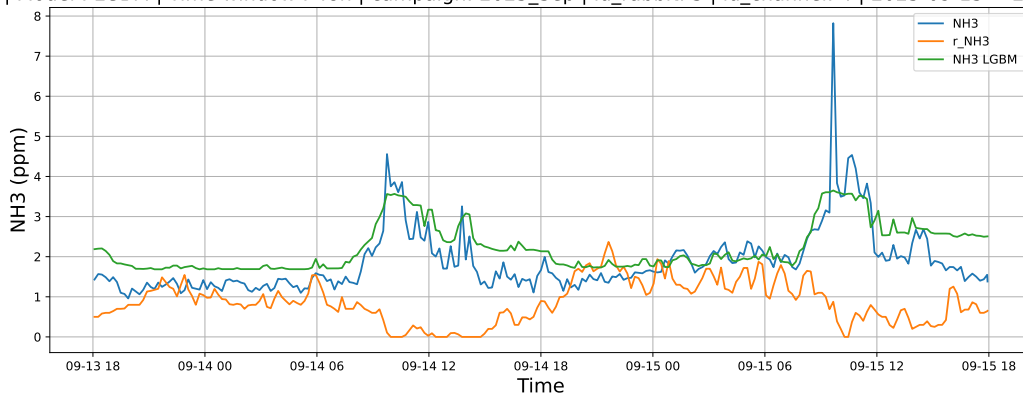


4.1.6.2 Time window : 48

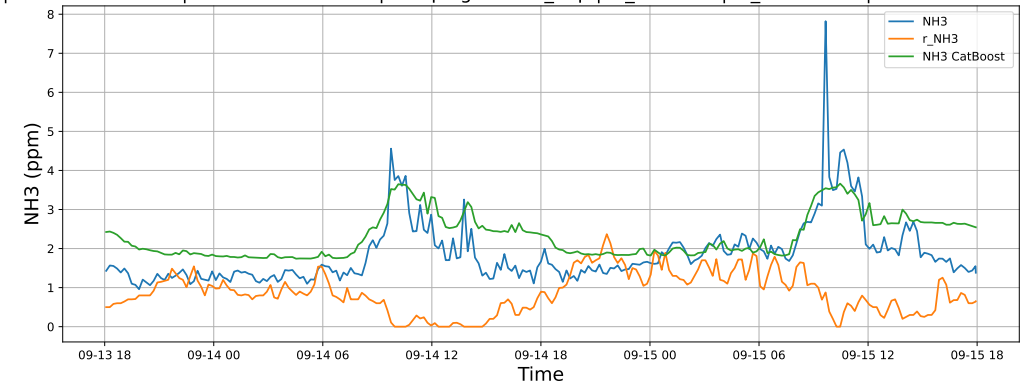
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

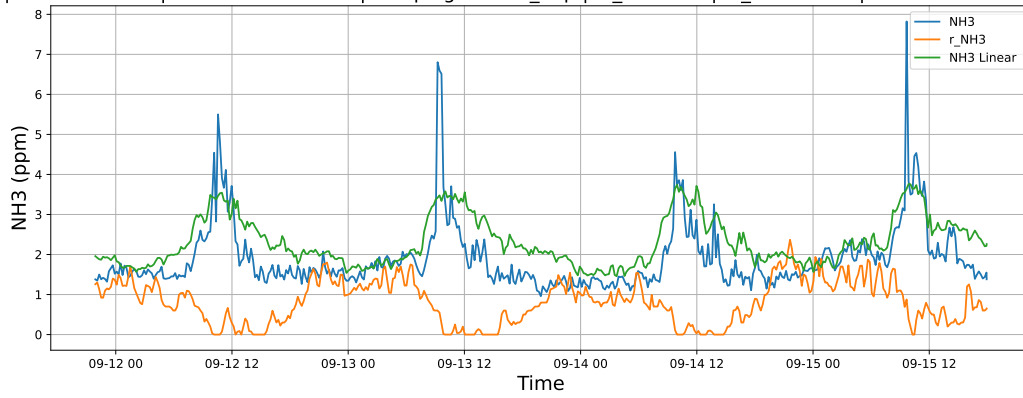


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

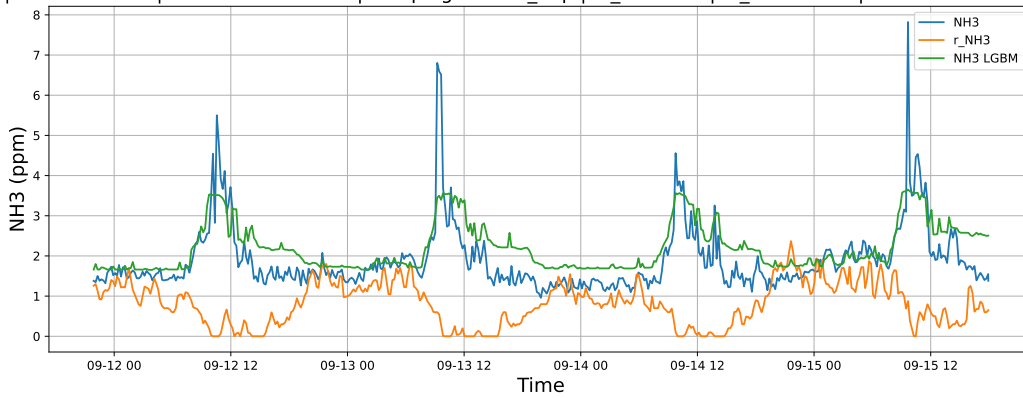


4.1.6.3 Time window : ALL

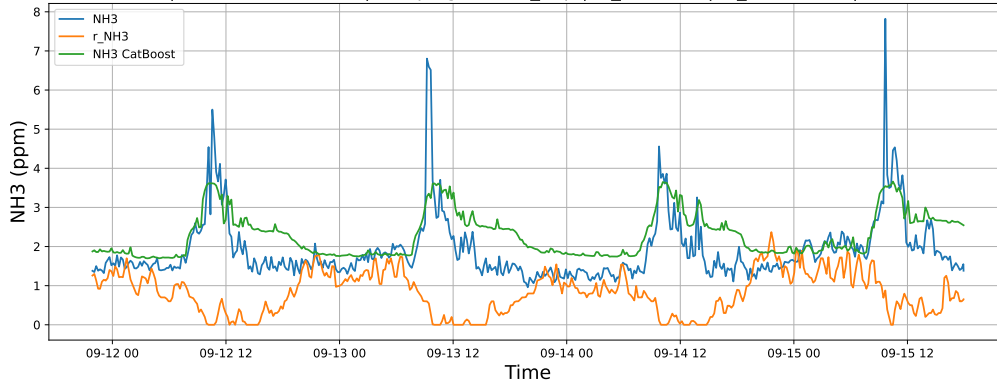
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



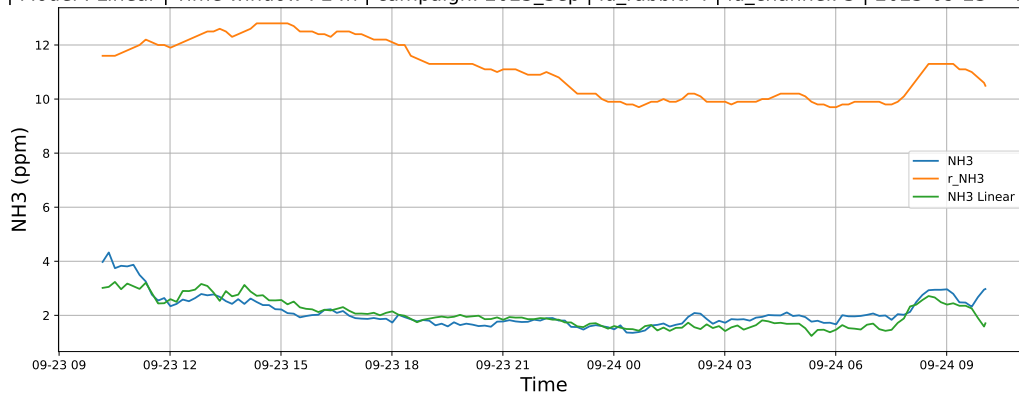
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



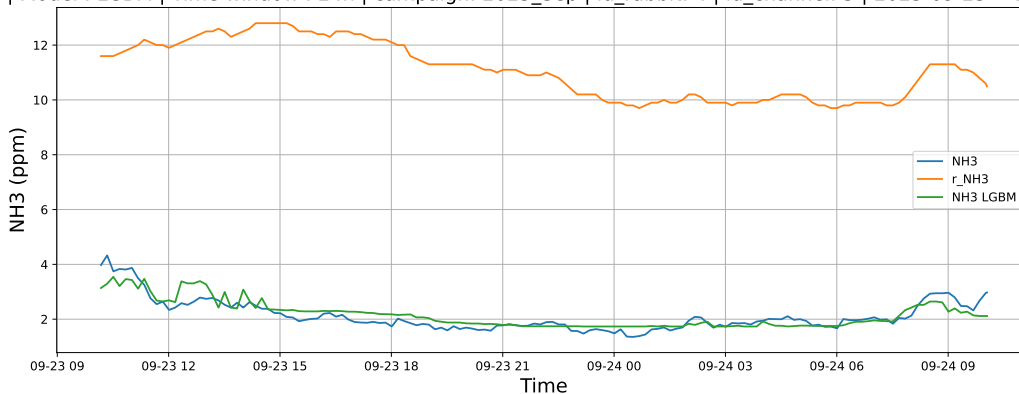
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

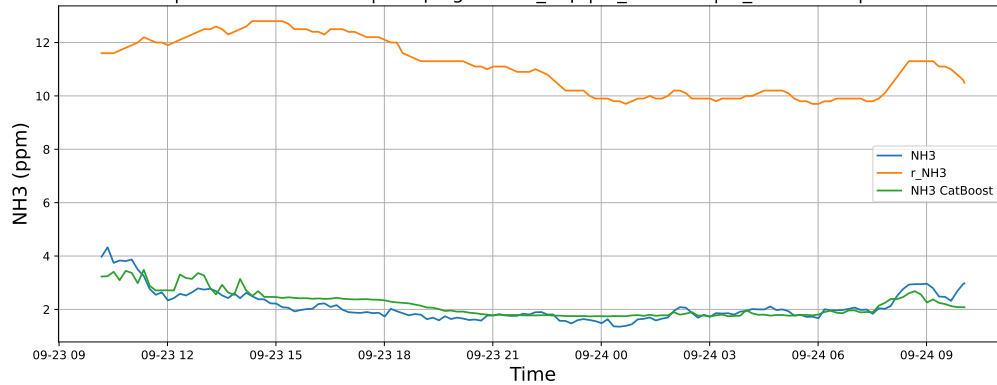
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

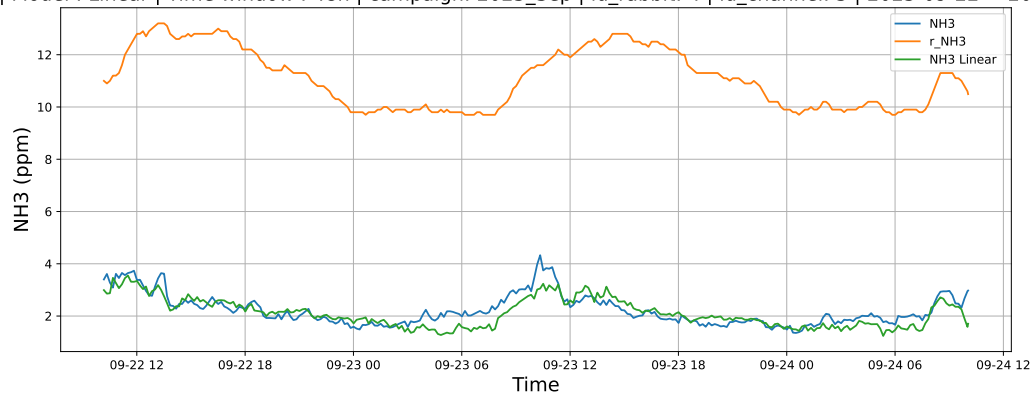


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

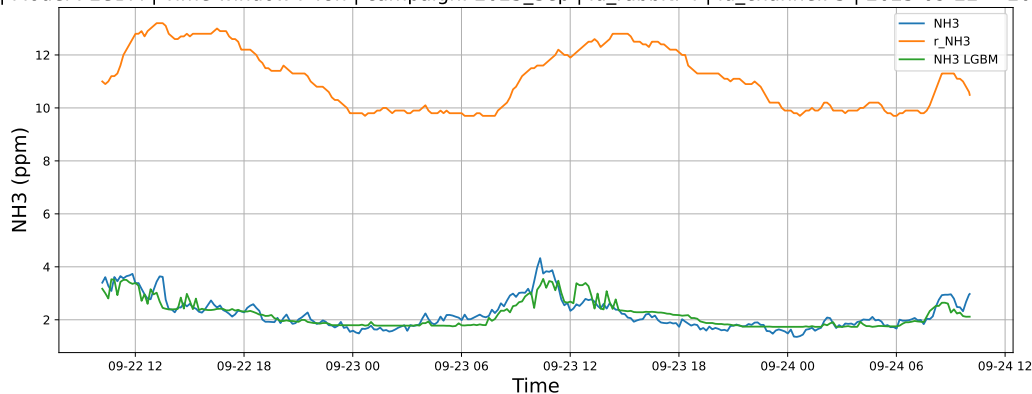


4.1.7.2 Time window : 48

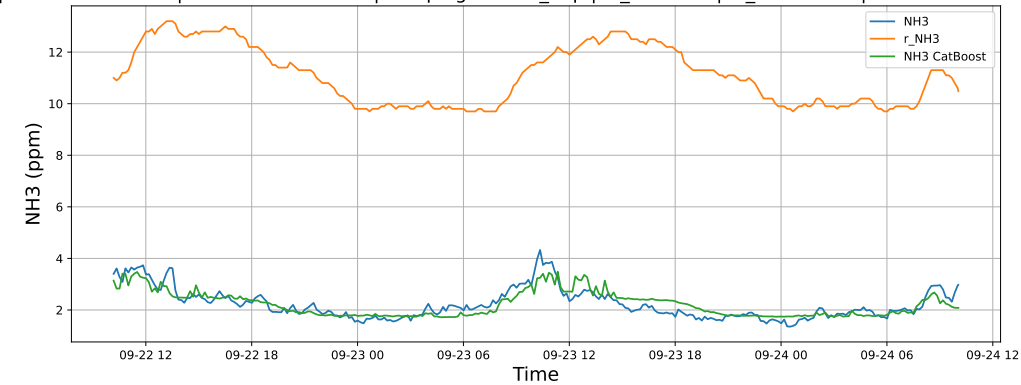
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

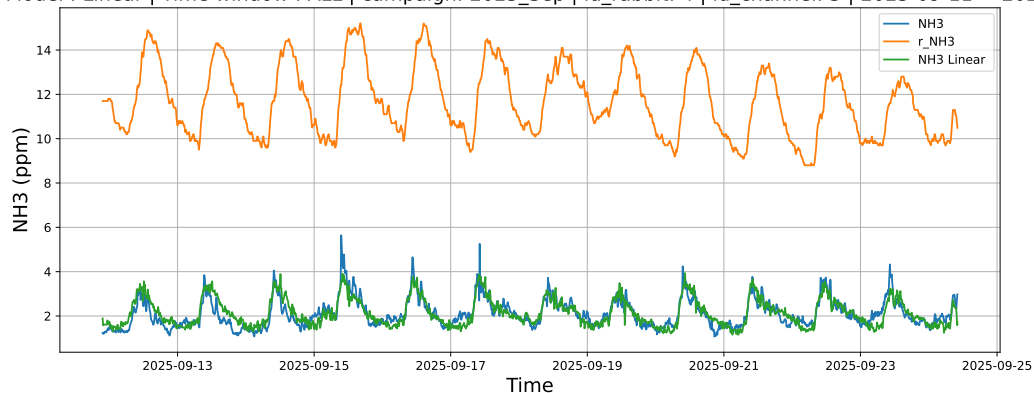


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

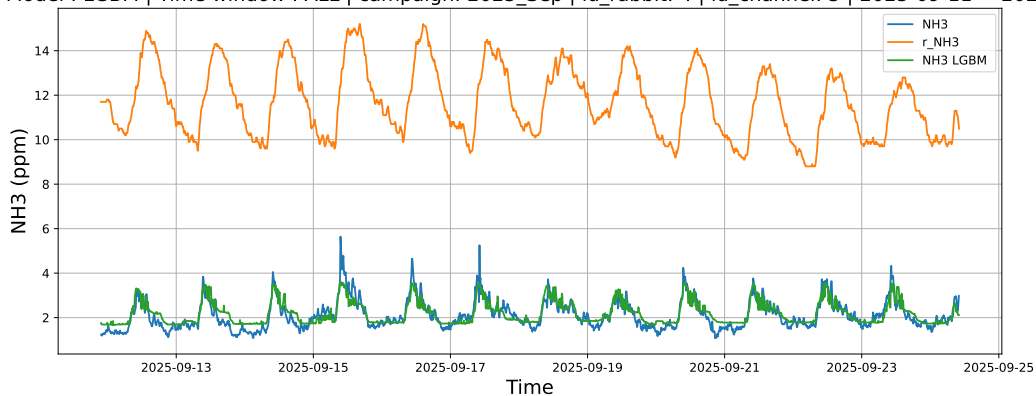


4.1.7.3 Time window : ALL

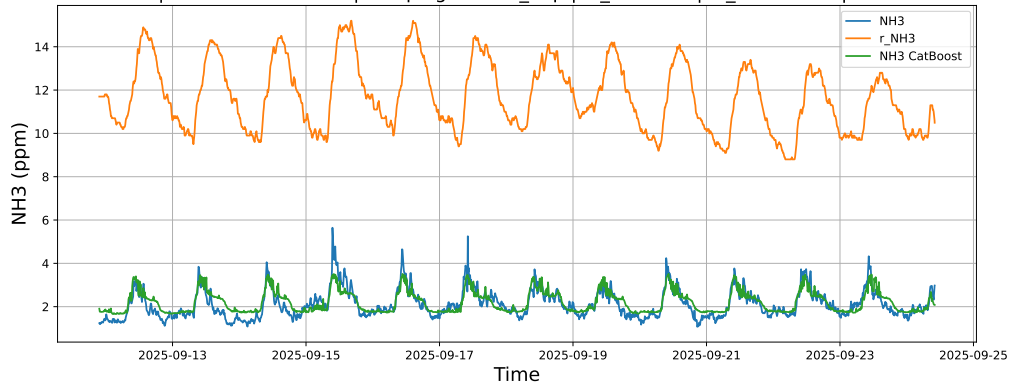
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



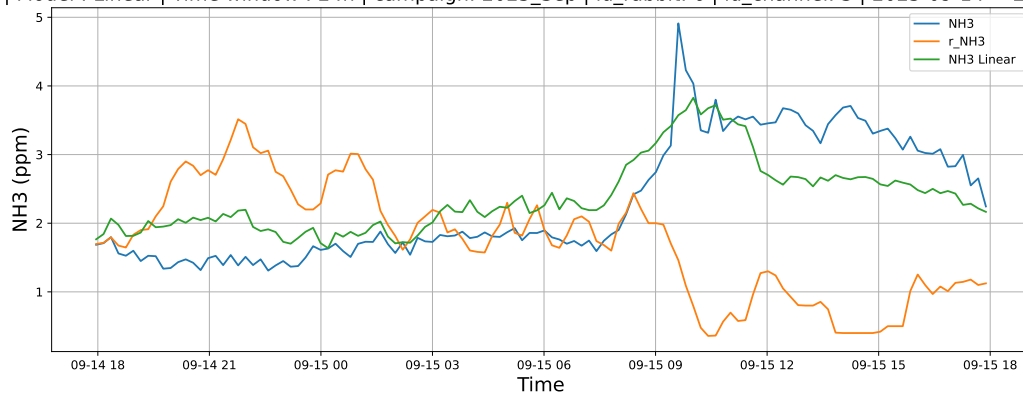
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



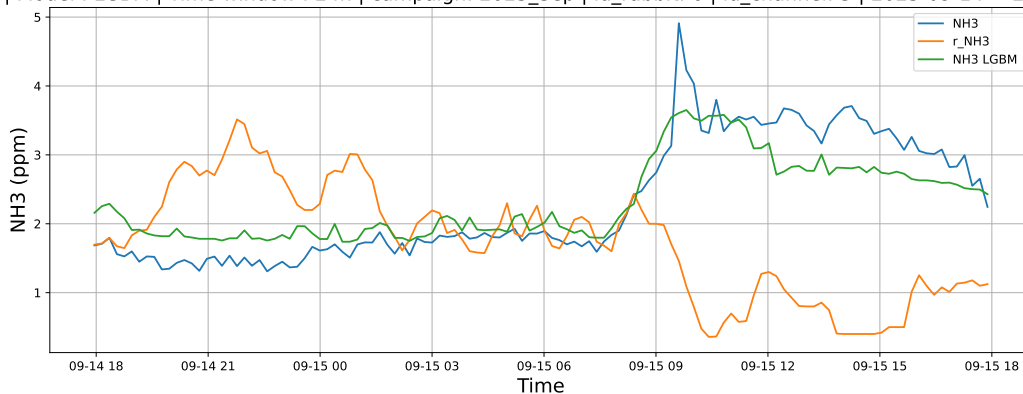
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

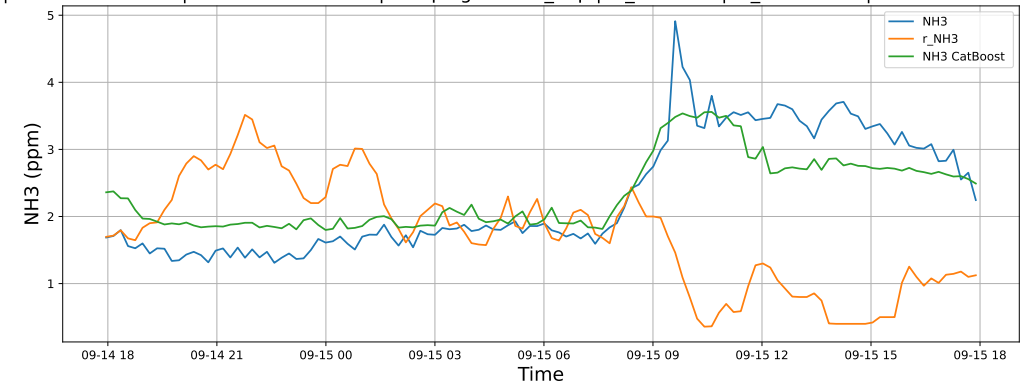
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

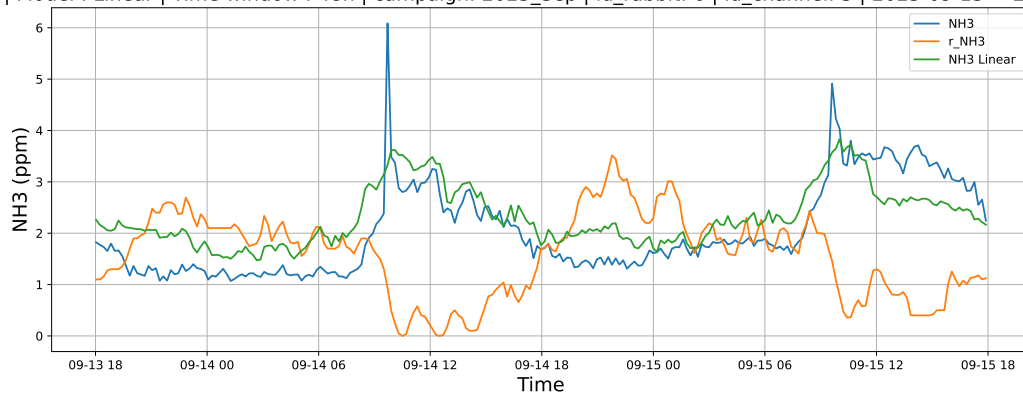


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

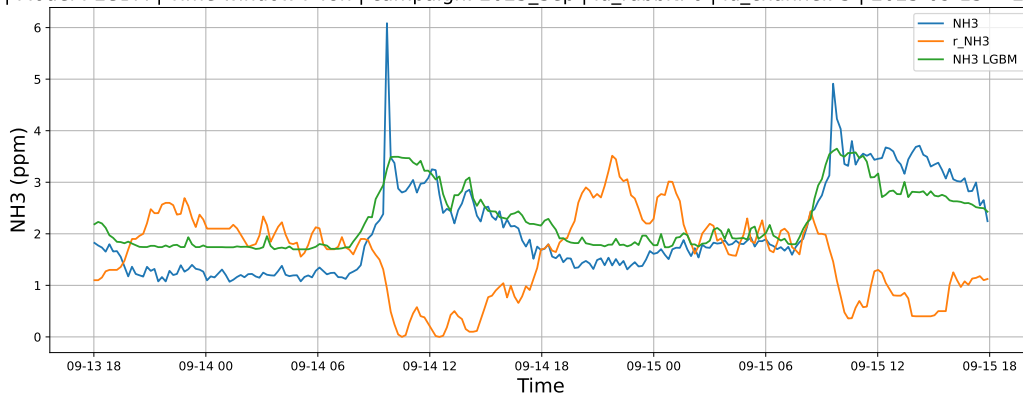


4.1.8.2 Time window : 48

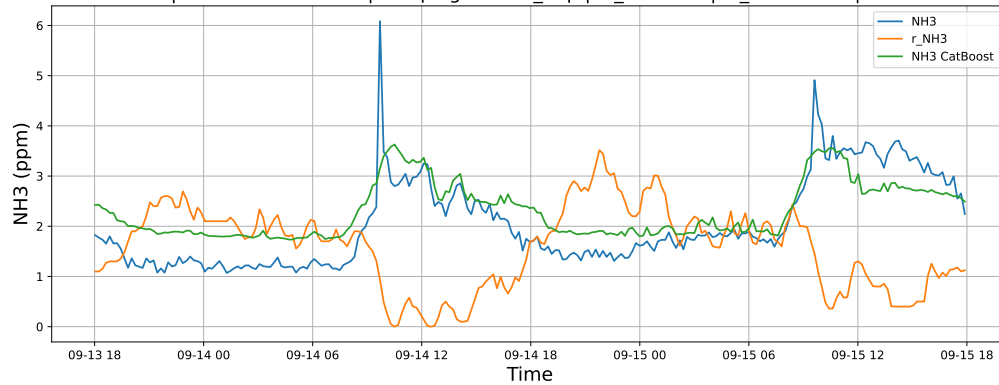
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

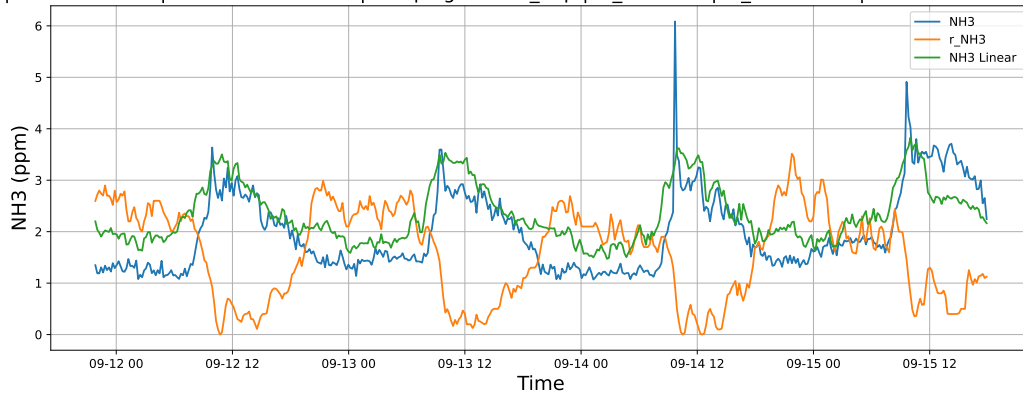


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

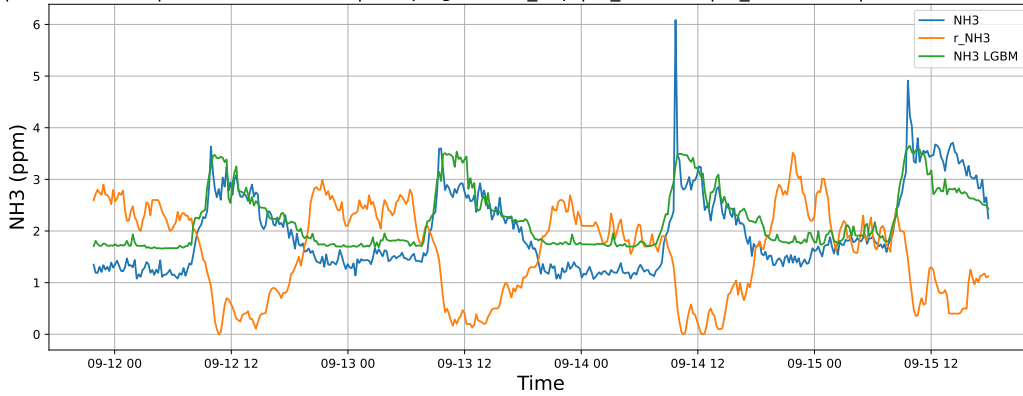


4.1.8.3 Time window : ALL

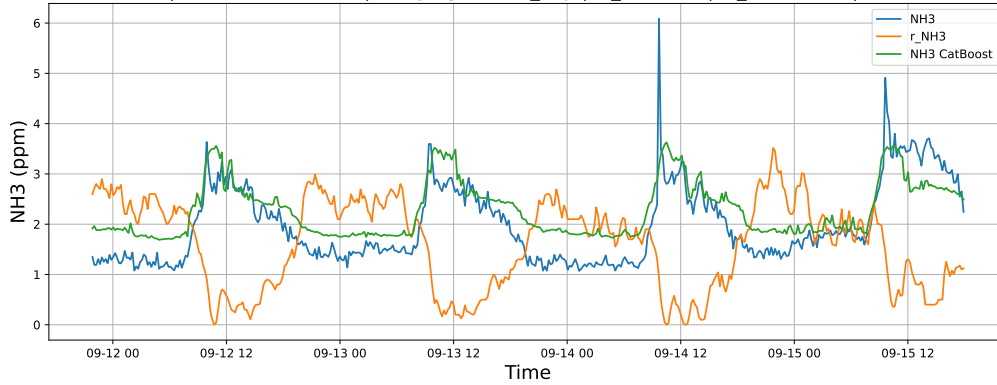
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

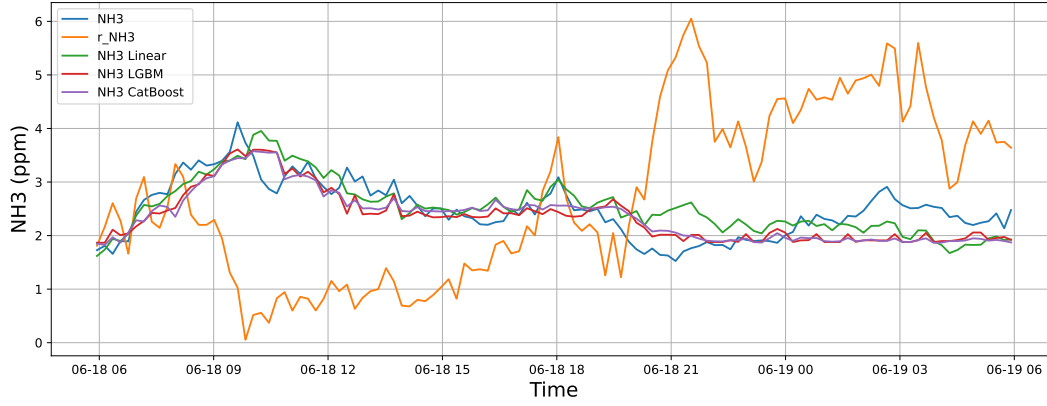


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

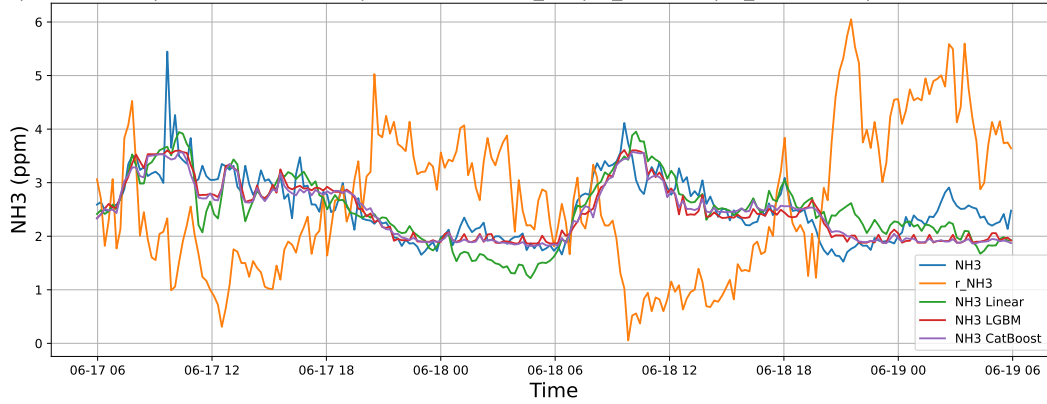
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



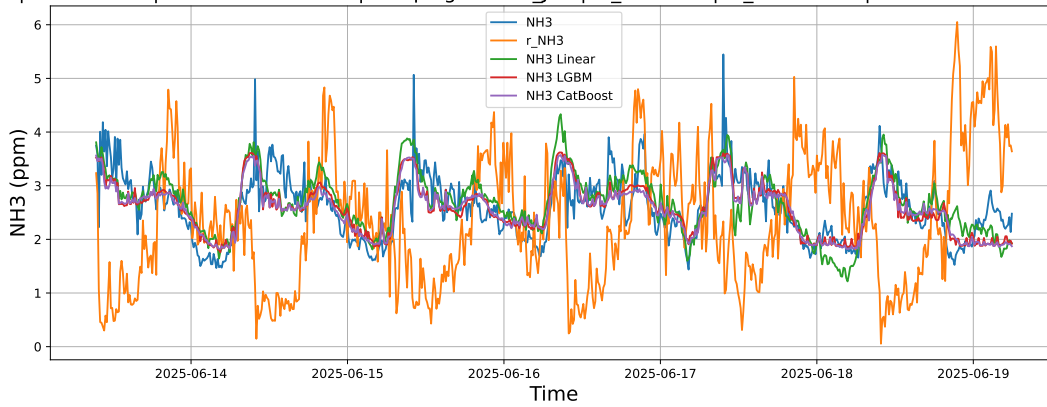
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

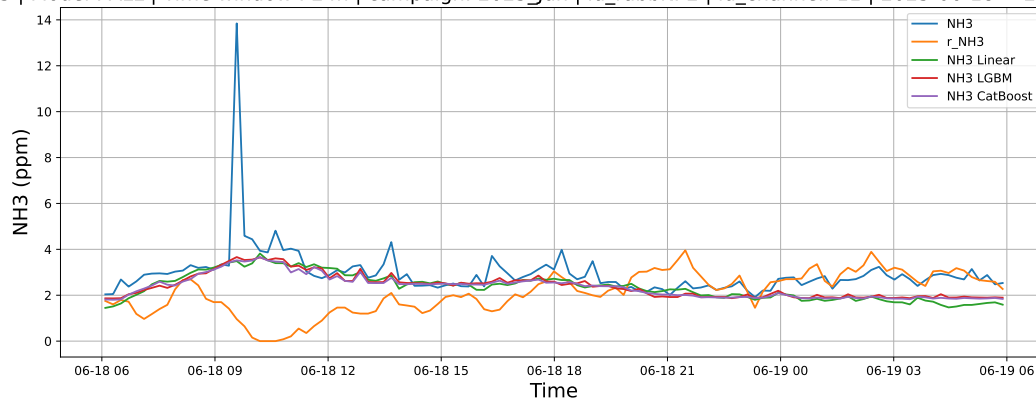
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

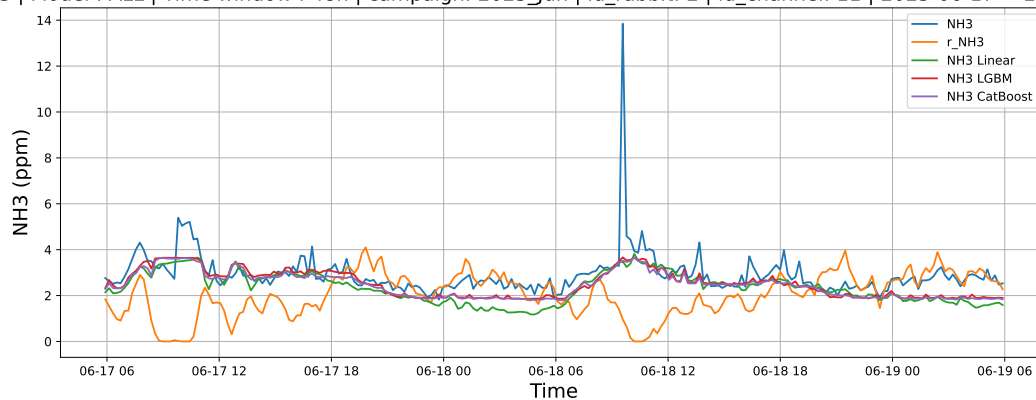
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



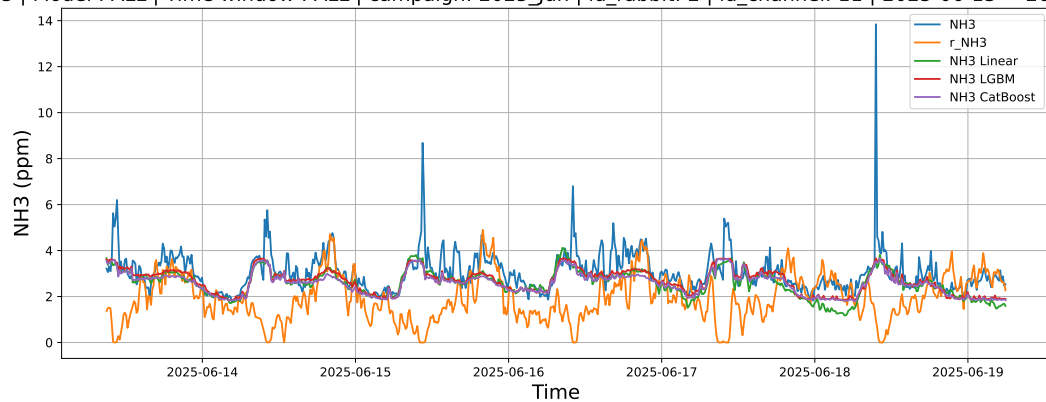
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

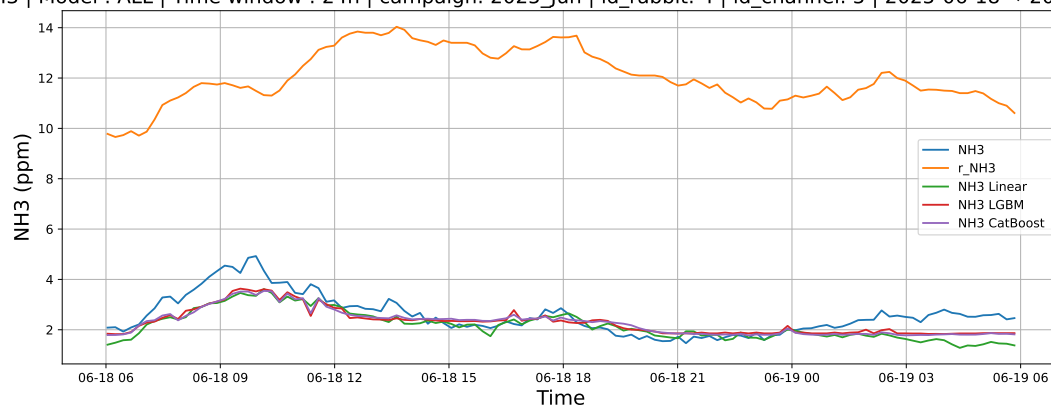
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

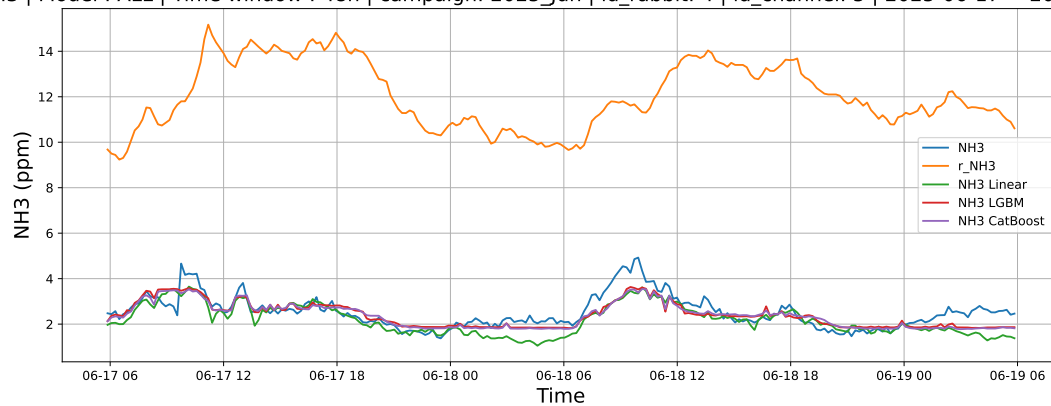
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



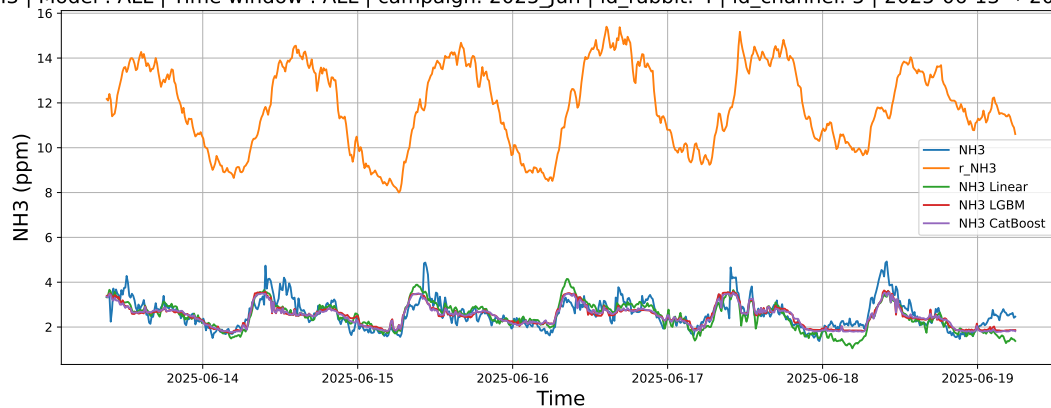
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

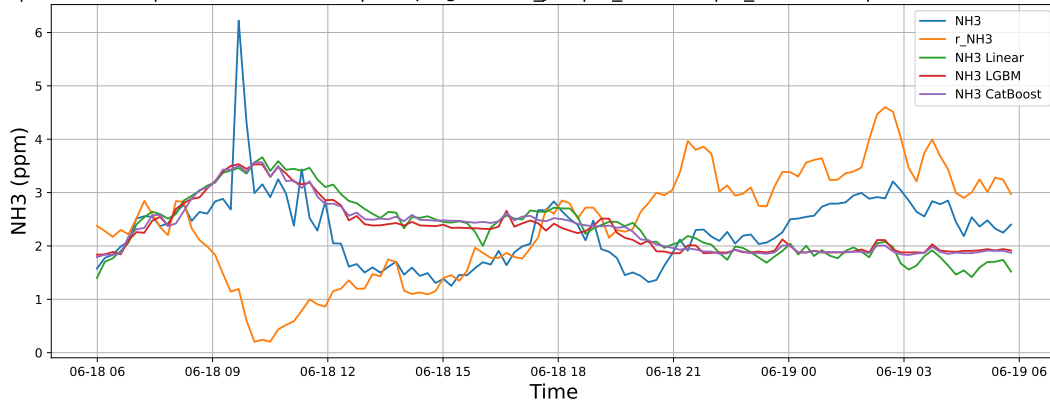
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

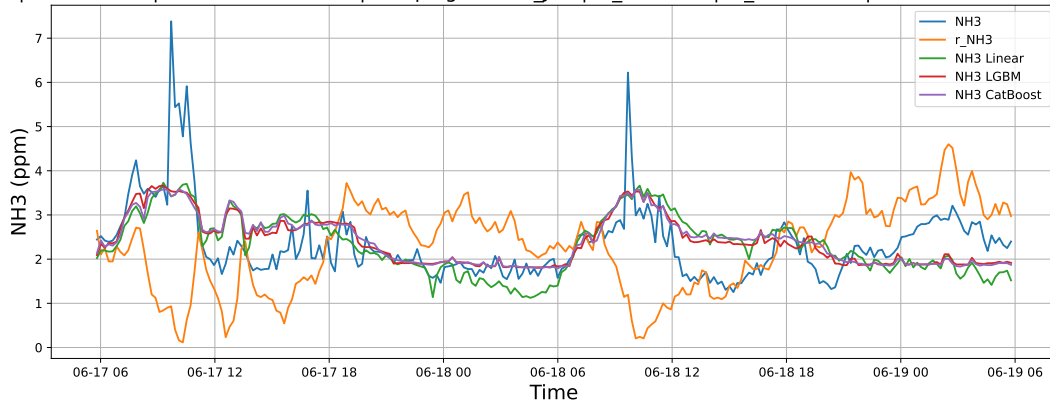
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



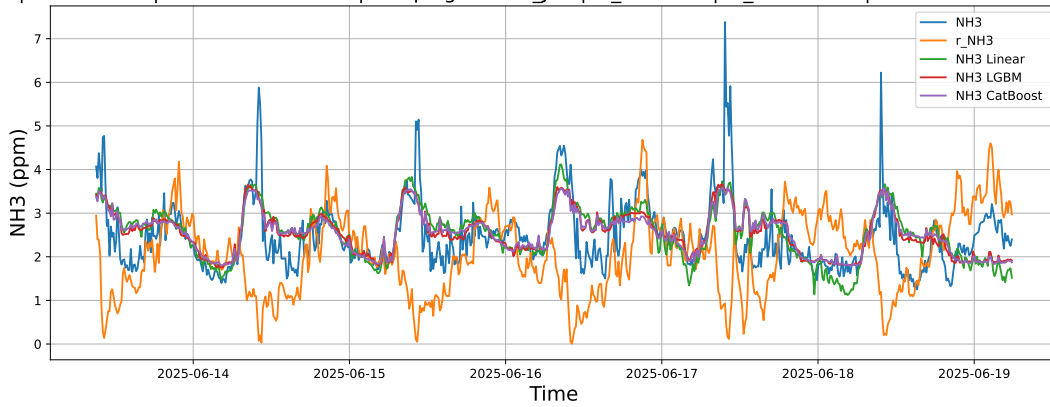
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

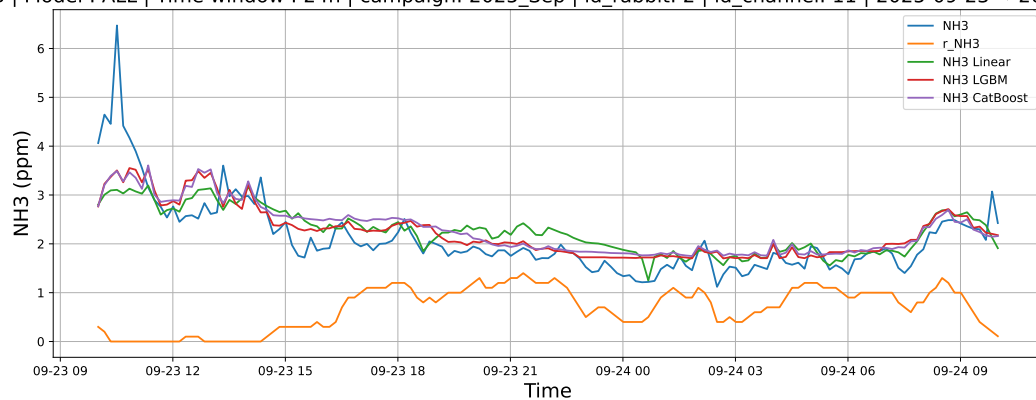
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

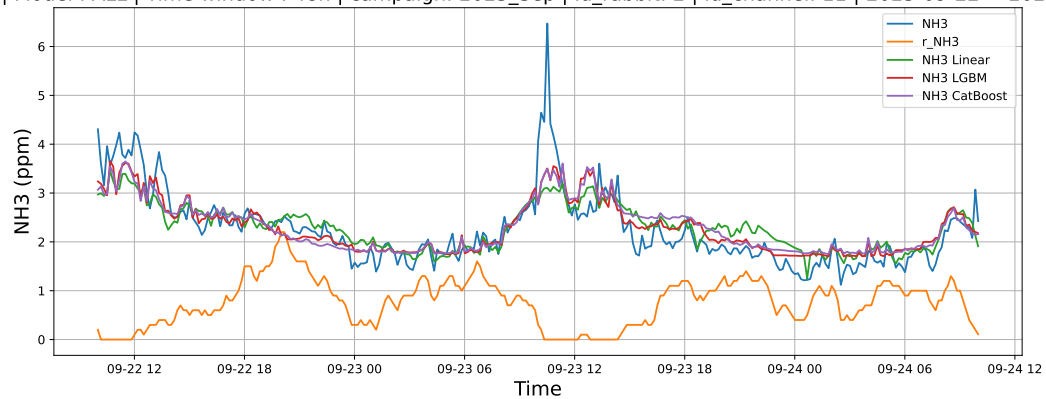
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



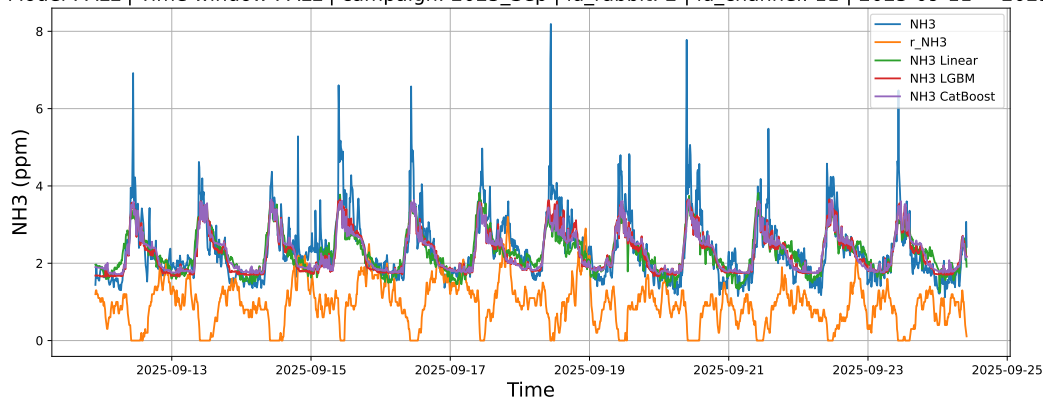
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

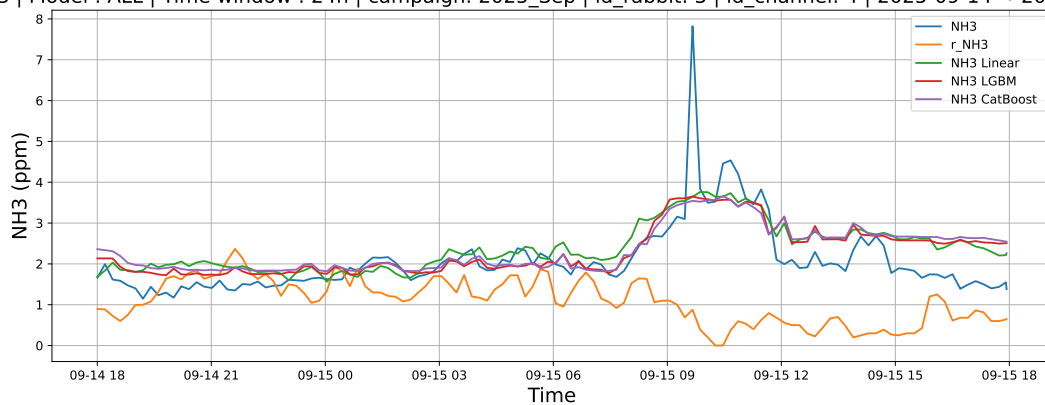
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

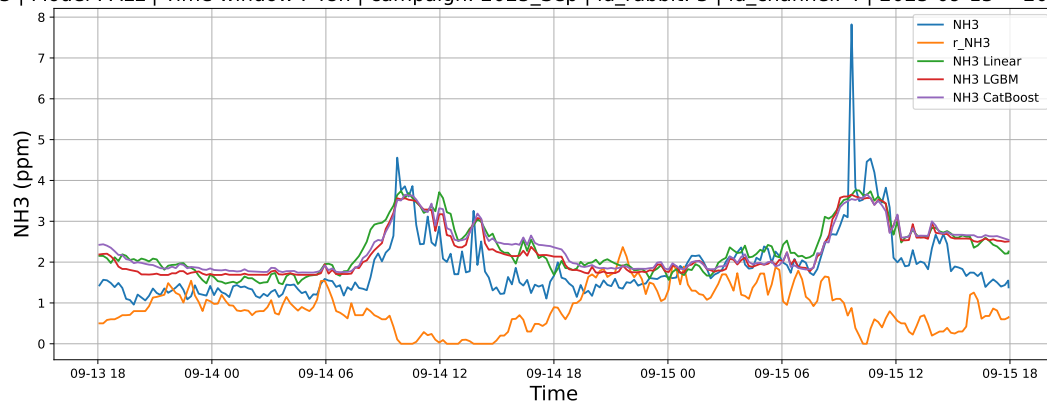
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



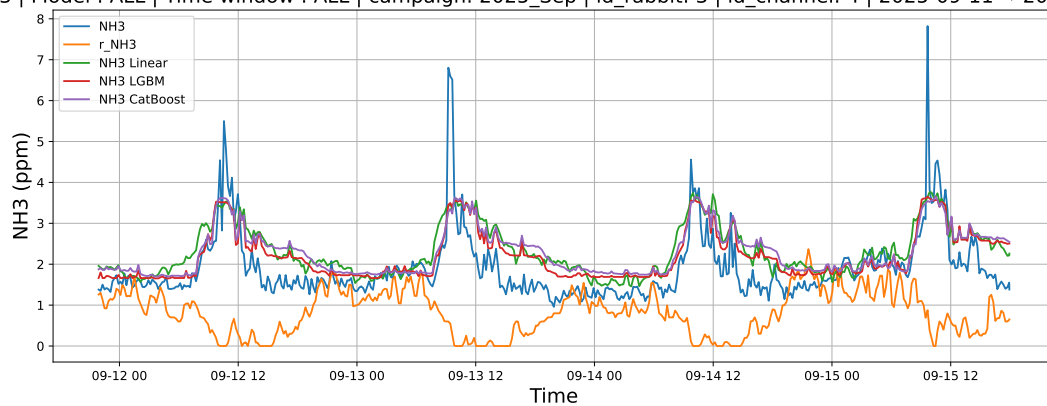
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

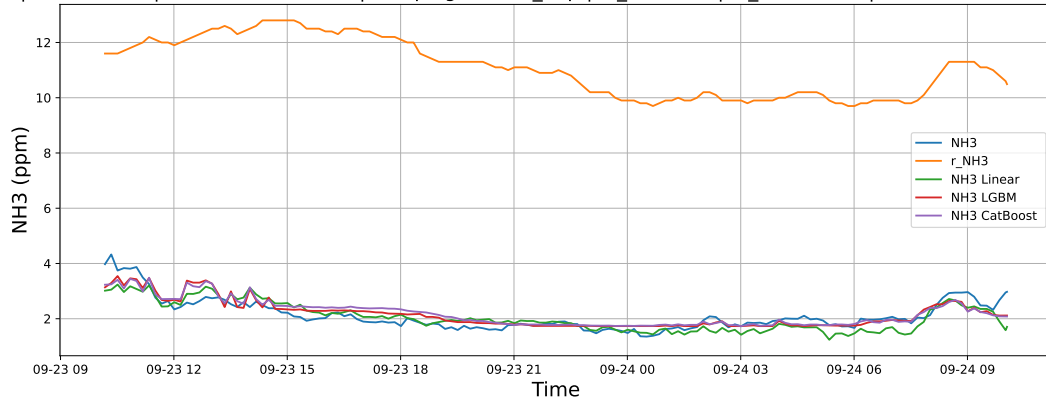
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

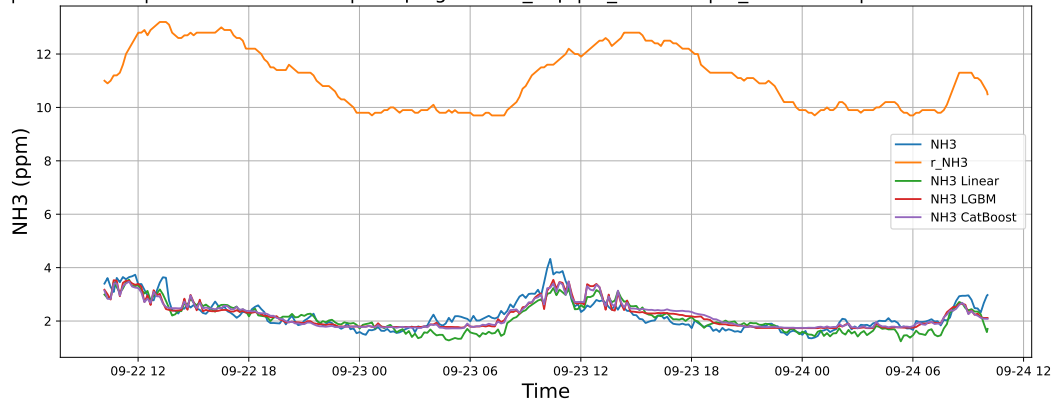
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



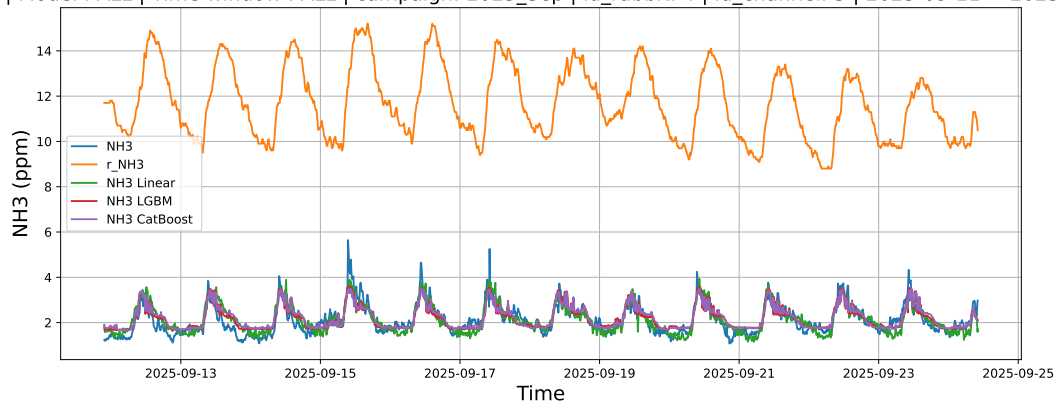
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

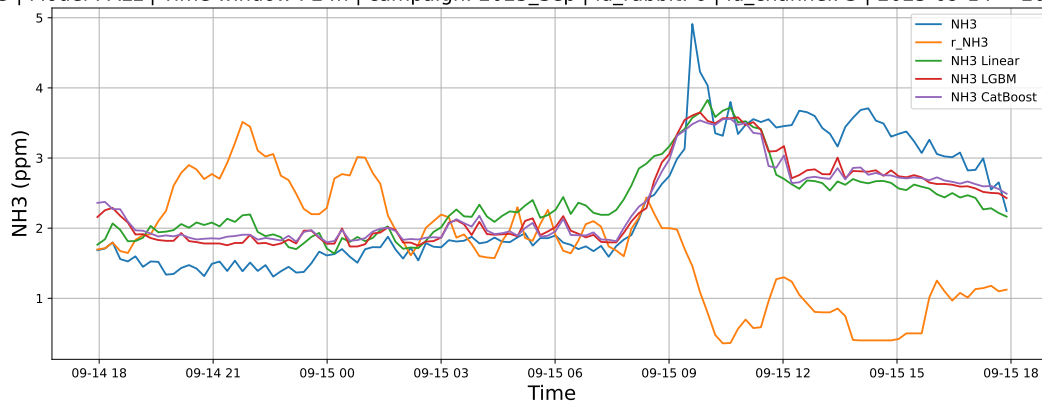
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

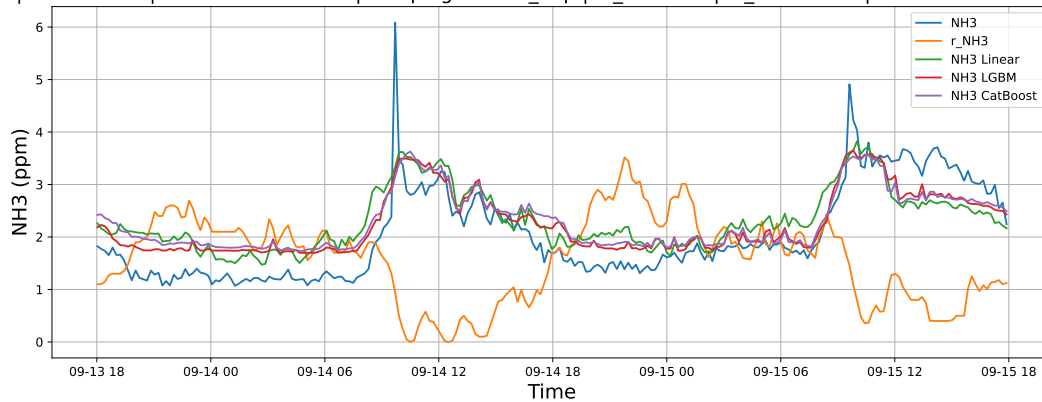
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

